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MIL-M-28787/508A
23 MARCH 1994
SUPERSEDING
MIL-M-28787/508
21 APRIL 1989

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

CENTRAL PROCESSOR SEQUENCER MODULE, KEY CODE BWAC

*This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the modules described herein shall consist of this specification and the latest issue of MIL-M-28787.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic Central Processor (CP) sequencer module.

1.2 Classification. This specification covers two classes of modules. Classes I and II as specified in MIL-STD-1389.

* Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Code 603A, Bldg. 2917, SHARP Project Office, NAVSURFWARCENDIV, 300 Highway 361, Crane, IN 47522-5001, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. The following specification and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

*(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Qualification. Modules furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list by the time of award of contract (see 4.5).

3.2 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.3 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.4 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 12.580 failures per million operating hours for class I or 33.143 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389.

3.5 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I or class II as specified in MIL-STD-1389.

3.5.1 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in class I environments, the failure rate for class I shall be used for reliability prediction.

3.6 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.6.1 Key code. The module key code shall be BWAC.

3.6.2 Size. The module size shall be 2A Format B.

3.6.3 Weight. The module shall have a maximum weight of 112 grams.

3.6.4 Contact pins. The module shall have 100 contact pins positioned 1 through 50 in one row and 51 through 100 in the other row.

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TABLE I. Module electrical requirements.

	Test cond	Parameter 5/	Ref test table	Test type 1/ 3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
					Min	Max	Min	Max	Min	Max	
*	A	Isolation	II	S	10	INF	10	INF	10	INF	MOhm
		End-of-life	II	S	1	INF	1	INF	1	INF	MOhm
	B1-B6	Continuity	III	X	0	1	0	1	0	1	Ohm
	C	Vcc stress	IV	S	-------Module shall pass all remaining tests----------						
	D1,D2	Ibd	V	S	0.0	1.0	0.0	1.0	0.0	1.0	mA
	D3-D5	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
	D6	Ibd	V	S	0.0	2.3	0.0	2.3	0.0	2.3	mA
	D7	Ibd	V	S	0.0	1.0	0.0	1.0	0.0	1.0	mA
	D8	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
	D9	Ibd	V	S	0.0	1.0	0.0	1.0	0.0	1.0	mA
	D10,D11	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
	D12	Ibd	V	S	0.0	1.1	0.0	1.1	0.0	1.1	mA
	D13	Ibd	V	S	-0.02	2.12	-0.02	2.12	-0.02	2.12	mA
	D14	Ibd	V	S	0.0	1.1	0.0	1.1	0.0	1.1	mA
	D15	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
	D16	Ibd	V	S	-0.02	2.12	-0.02	2.12	-0.02	2.12	mA
	D17	Ibd	V	S	0.0	2.0	0.0	2.0	0.0	2.0	mA
	D18	Ibd	V	S	0.0	1.0	0.0	1.0	0.0	1.0	mA
	D19	Ibd	V	S	0.0	3.0	0.0	3.0	0.0	3.0	mA
	E1	Ios	VI	S	-100.0	-40.0	-100.0	-40.0	-100.0	-40.0	mA
	E2,E3	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
	E4	Ios	VI	S	-130.0	-15.0	-130.0	-15.0	-130.0	-15.0	mA
	E5,E6	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
*	E7-E10	Ios	VI	S	-160.0	-12.0	-160.0	-12.0	-160.0	-12.0	mA
	E11,E12	Ios	VI	S	-85.0	-30.0	-85.0	-30.0	-85.0	-30.0	mA
	E13	Ios	VI	S	-100.0	-40.0	-100.0	-40.0	-100.0	-40.0	mA
	E14,E15	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/ - 3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
E16-E20	Ios	VI	S	-260.0	-30.0	-260.0	-30.0	-260.0	-30.0	mA
E21	Ios	VI	S	-100.0	-40.0	-100.0	-40.0	-100.0	-40.0	mA
E22-E36	Ios	VI	S	-260.0	-30.0	-260.0	-30.0	-260.0	-30.0	mA
E37-E46	Ios	VI	S	-85.0	-30.0	-85.0	-30.0	-85.0	-30.0	mA
E47-E58	Ios	VI	S	-260.0	-30.0	-260.0	-30.0	-260.0	-30.0	mA
E59,E60	Ios	VI	S	-130.0	-30.0	-130.0	-30.0	-130.0	-30.0	mA
E61	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E62-E73	Ios	VI	S	-260.0	-30.0	-260.0	-30.0	-260.0	-30.0	mA
F1,F2	Iih	VII	X	-2.0	0.05	-2.0	0.05	-2.0	0.05	mA
F3-F5	Iih	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F6	Iih	VII	X	-5.8	0.16	-5.8	0.16	-5.8	0.16	mA
F7	Iih	VII	X	-2.0	0.05	-2.0	0.05	-2.0	0.05	mA
F8,F9	Iih	VII	X	-0.8	0.24	-0.8	0.24	-0.8	0.24	mA
F10	Iih	VII	X	-3.65	0.21	-3.65	0.21	-3.65	0.21	mA
F11	Iih	VII	X	-3.67	0.23	-3.67	0.23	-3.67	0.23	mA
F12	Iih	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F13	Iih	VII	X	-2.0	0.05	-2.0	0.05	-2.0	0.05	mA
F14	Iih	VII	X	-0.4	0.02	-0.4	0.02	-0.4	0.02	mA
F15	Iih	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F16-F25	Iih	VII	X	-3.65	0.21	-3.65	0.21	-3.65	0.21	mA
F26	Iih	VII	X	-2.4	0.07	-2.4	0.07	-2.4	0.07	mA
F27	Iih	VII	X	-3.11	-0.52	-3.11	-0.52	-3.11	-0.52	mA
F28	Iih	VII	X	-2.6	0.07	-2.6	0.07	-2.6	0.07	mA
F29	Iih	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F30	Iih	VII	X	-5.75	-0.49	-5.75	-0.49	-5.75	-0.49	mA
F31	Iih	VII	X	-4.0	0.1	-4.0	0.1	-4.0	0.1	mA
F32	Iih	VII	X	-2.0	0.05	-2.0	0.05	-2.0	0.05	mA
F33	Iih	VII	X	-6.0	0.15	-6.0	0.15	-6.0	0.15	mA
G1,G2	Iii	VIII	X	-2.0	0.05	-2.0	0.05	-2.0	0.05	mA

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/ - 3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
G3-G5	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G6	Iil	VIII	X	-5.8	0.16	-5.8	0.16	-5.8	0.16	mA
G7	Iil	VIII	X	-2.0	0.05	-2.0	0.05	-2.0	0.05	mA
G8,G9	Iil	VIII	X	-0.8	0.24	-0.8	0.24	-0.8	0.24	mA
G10	Iil	VIII	X	-3.65	0.21	-3.65	0.21	-3.65	0.21	mA
G11	Iil	VIII	X	-3.67	0.23	-3.67	0.23	-3.67	0.23	mA
G12	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G13	Iil	VIII	X	-2.0	0.05	-2.0	0.05	-2.0	0.05	mA
G14	Iil	VIII	X	-0.4	0.02	-0.4	0.02	-0.4	0.02	mA
G15	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G16-G25	Iil	VIII	X	-3.65	0.21	-3.65	0.21	-3.65	0.21	mA
G26	Iil	VIII	X	-2.4	0.07	-2.4	0.07	-2.4	0.07	mA
G27	Iil	VIII	X	-3.11	-0.52	-3.11	-0.52	-3.11	-0.52	mA
G28	Iil	VIII	X	-2.6	0.07	-2.6	0.07	-2.6	0.07	mA
G29	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G30	Iil	VIII	X	-5.75	-0.49	-5.75	-0.49	-5.75	-0.49	mA
G31	Iil	VIII	X	-4.0	0.1	-4.0	0.1	-4.0	0.1	mA
G32	Iil	VIII	X	-2.0	0.05	-2.0	0.05	-2.0	0.05	mA
G33	Iil	VIII	X	-6.0	0.15	-6.0	0.15	-6.0	0.15	mA
H1-H73	Voh	IX	X	2.4	4.5	2.4	4.5	2.4	4.5	V
J1-J74	Vol	X	X	0.0	0.5	0.0	0.5	0.0	0.5	V
L1-L9	Vic	XI	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L10	Vic	XI	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L11-L17	Vic	XI	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
M1	PSFC	XII	S	4.9	9.1	4.9	9.1	4.9	9.1	µF
M2	PSFC	XII	S	5.6	10.4	5.6	10.4	5.6	10.4	µF
* N	Icc	XIII	X	630	2544	630	2544	630	2544	mA

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/ - 3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
P1,P2	Functional	XIV	X	See table XIV						
Q1,Q2	Dynamic	XV	S	See table XV						
Q3,Q4	Dynamic	XV	Q	See table XV						
R1-R5	Ci/o	XVII	Q	2	50	2	50	2	50	pF
R6	Ci/o	XVII	Q	10	100	10	100	10	100	pF
R7-R11	Ci/o	XVII	Q	2	50	2	50	2	50	pF
R12	Ci/o	XVII	Q	4	75	4	75	4	75	pF
R13	Ci/o	XVII	Q	8	175	8	175	8	175	pF
R14	Ci/o	XVII	Q	4	100	4	100	4	100	pF
R15	Ci/o	XVII	Q	2	50	2	50	2	50	pF
R16	Ci/o	XVII	Q	8	125	8	125	8	125	pF
R17	Ci/o	XVII	Q	4	75	4	75	4	75	pF
R18	Ci/o	XVII	Q	2	75	2	75	2	75	pF
R19	Ci/o	XVII	Q	6	100	6	100	6	100	pF
R20	Ci/o	XVII	Q	16	321	16	321	16	321	pF
R21	Ci/o	XVII	Q	16	312	16	312	16	312	pF
R22	Ci/o	XVII	Q	14	364	14	364	14	364	pF
R23	Ci/o	XVII	Q	14	200	14	200	14	200	pF
R24	Ci/o	XVII	Q	14	273	14	273	14	273	pF
R25	Ci/o	XVII	Q	14	263	14	263	14	263	pF
R26	Ci/o	XVII	Q	14	262	14	262	14	262	pF
R27	Ci/o	XVII	Q	14	254	14	254	14	254	pF
R28	Ci/o	XVII	Q	14	261	14	261	14	261	pF
R29	Ci/o	XVII	Q	14	283	14	283	14	283	pF
R30	Ci/o	XVII	Q	14	279	14	279	14	279	pF
R31	Ci/o	XVII	Q	14	328	14	328	14	328	pF
R32,R33	Ci/o	XVII	Q	2	50	2	50	2	50	pF
R34,R35	Ci/o	XVII	Q	4	75	4	75	4	75	pF
R36	Ci/o	XVII	Q	4	100	4	100	4	100	pF
R37	Ci/o	XVII	Q	12	204	12	204	12	204	pF
R38	Ci/o	XVII	Q	10	150	10	150	10	150	pF
R39	Ci/o	XVII	Q	8	175	8	175	8	175	pF

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/ - 3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
R40	Ci/o	XVII	Q	8	200	8	200	8	200	pF
R41-R47	Ci/o	XVII	Q	2	50	2	50	2	50	pF
R48-R64	Ci/o	XVII	Q	4	75	4	75	4	75	pF
R65	Ci/o	XVII	Q	2	75	2	75	2	75	pF
R66	Ci/o	XVII	Q	2	50	2	50	2	50	pF
R67	Ci/o	XVII	Q	2	75	2	75	2	75	pF
R68	Ci/o	XVII	Q	4	150	4	150	4	150	pF
R69	Ci/o	XVII	Q	6	200	6	200	6	200	pF
R70	Ci/o	XVII	Q	4	75	4	75	4	75	pF
R71	Ci/o	XVII	Q	2	75	2	75	2	75	pF
R72-R75	Ci/o	XVII	Q	4	75	4	75	4	75	pF
R76	Ci/o	XVII	Q	2	75	2	75	2	75	pF
T1,T2	PSTA	XVIII	Q	0	400	0	400	0	400	mV

- See footnotes at end of table.

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1/ "X" parameters shall be 100 percent tested during the module acceptance test.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified in the test table, limits are for initial and end-of-life.

5/ Abbreviations used in table I are defined in 6.1.

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3.6.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.6.6 Depth. The module shall have no additional depth requirements.

3.6.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.6.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part or identifying number M28787/508-1 for class I and M28787/508-2 for class II. No marking shall interfere with heat dissipation from the fin (such as affixing a label which may act as a heat barrier).

3.7 Circuit pin configuration. The functional block diagram shall conform to figure 1. The pin assignments shall conform to figure 2.

3.8 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 in table I.

3.9 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits in table I (see 6.2).

3.10 Power supply voltage. Unless otherwise specified herein, the module shall be capable of conforming to the requirements in table I for a power supply voltage (Vcc) of 5.5 V dc.

3.11 Parallel outputs. Paralleling of outputs is not permissible.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

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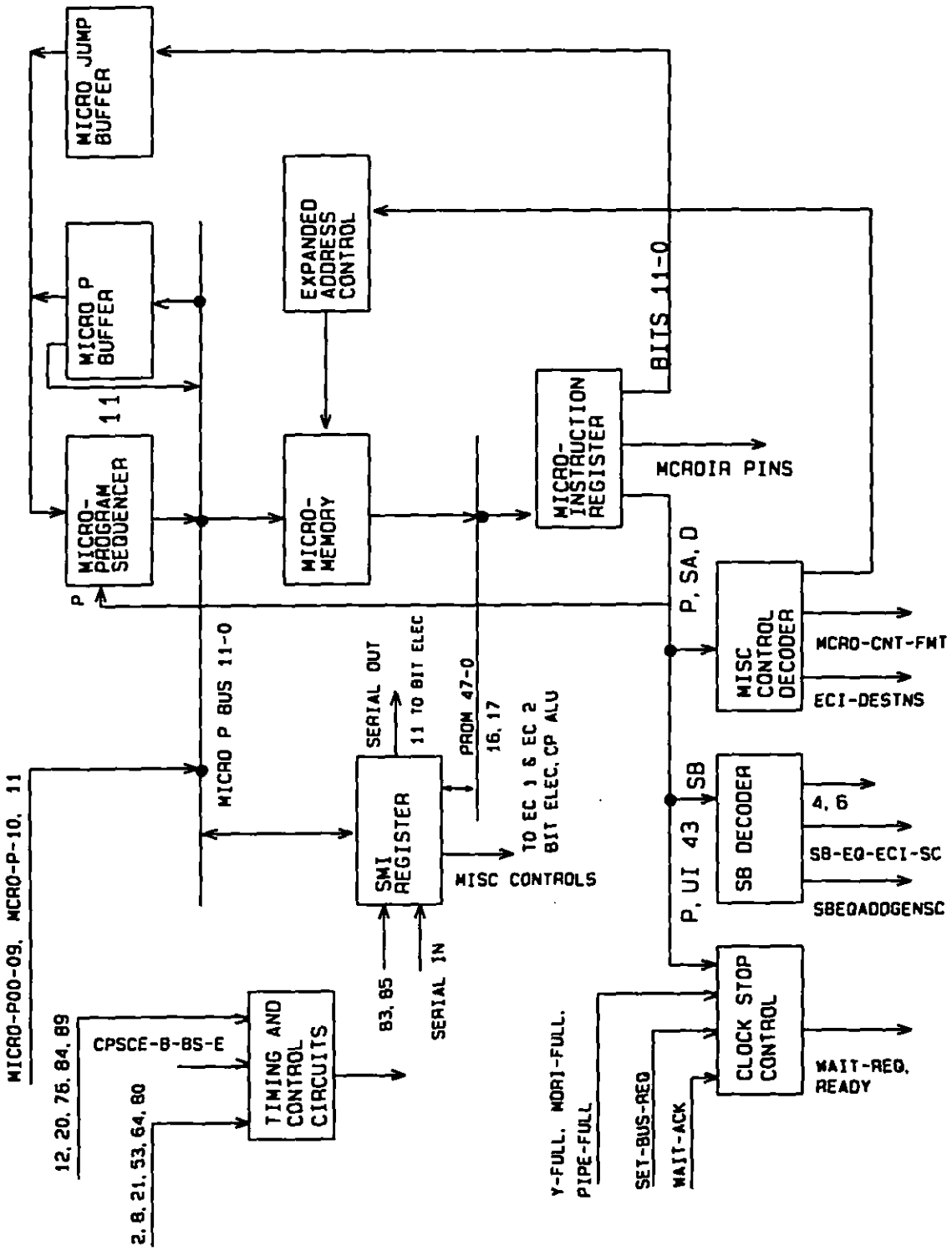


FIGURE 1. BWAC functional block diagram.

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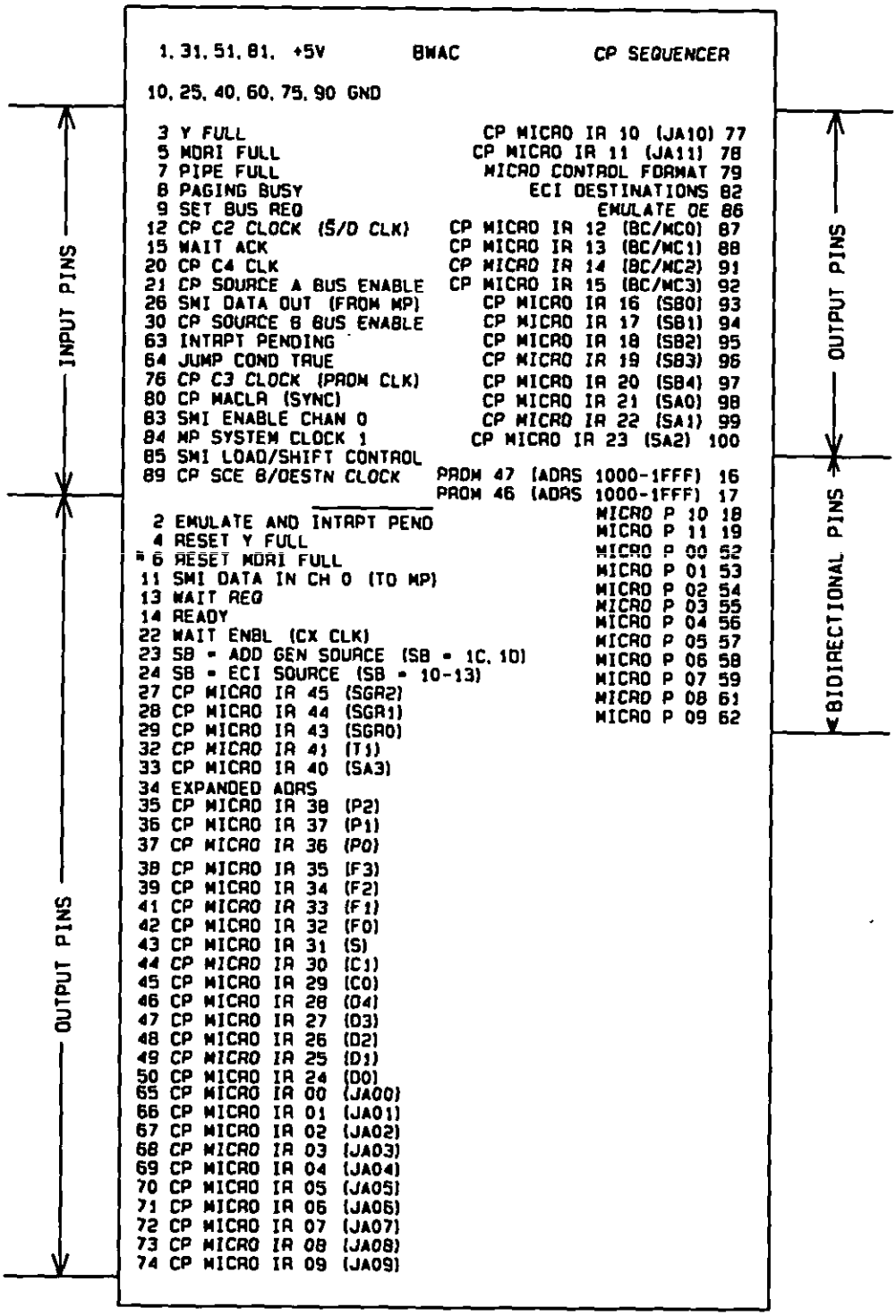


FIGURE 2. BWAC pin assignment.

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4.3 Test equipment. Test equipment specified on figures and in tables with a designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. Unless otherwise specified herein, all resistors shall be metal film and all capacitors shall be ceramic.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Quality Assurance Activity (SEM-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through T shall be performed in any order with the following exceptions. Test conditions A, C, D, E and L are stress tests and shall be performed only at 25 ± 5°C. If test conditions A, C, D, E or L are part of a sequence of tests, the order of testing shall be as follows: A, C, D, E, and L where applicable, followed by the remaining tests in the sequence.

4.4.3 Multiple testing of pins. Singular pins listed multiple times in a test table indicate multiple devices connected to the pins. The module shall be tested so that each device connected to a pin is exercised and tested.

4.5 Qualification testing. Initial qualification tests and periodic check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through T in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions B, F through J and M through Q shall be performed in the order specified in 4.4.2 using the initial limits for the applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 3.

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4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests in table I which are marked with an "X" in the test type column, and test condition M, shall be performed using the initial limits at 25 ± 5°C ambient.

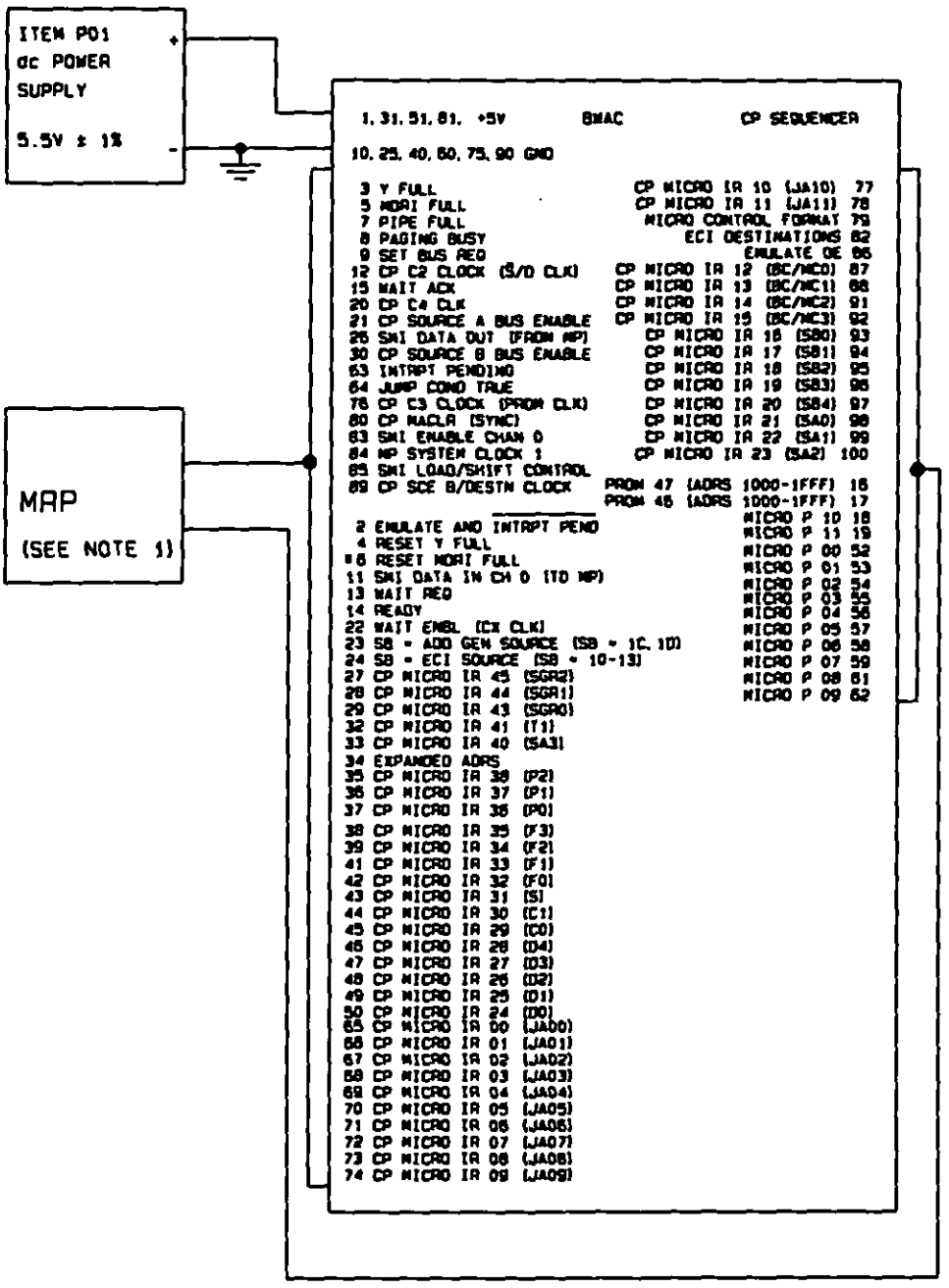
4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 5. Following the life test and while still at the maximum class temperature, electrical tests B, F through J and M through Q shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements in table I. Following the return of the module to 25 ± 5°C, electrical tests B, F through J and M through Q shall be performed in the order specified in 4.4.2 using the 25 ± 5°C ambient end-of-life requirements in table I.

4.5.4 Thermal shock test. The thermal shock test shall be conducted in accordance with MIL-M-28787. The module shall be unpowered during this test. Following the thermal shock test, the electrical tests in table I which are marked with an "X" in the test type column, shall be performed using the initial limits at 25 ± 5°C ambient.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

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NOTE:

1. Military Reconfigurable Processor (MRP) or emulator capable of reproducing system backpanel characteristics.

FIGURE 3. BWAC operating temperature and component temperature test circuit.

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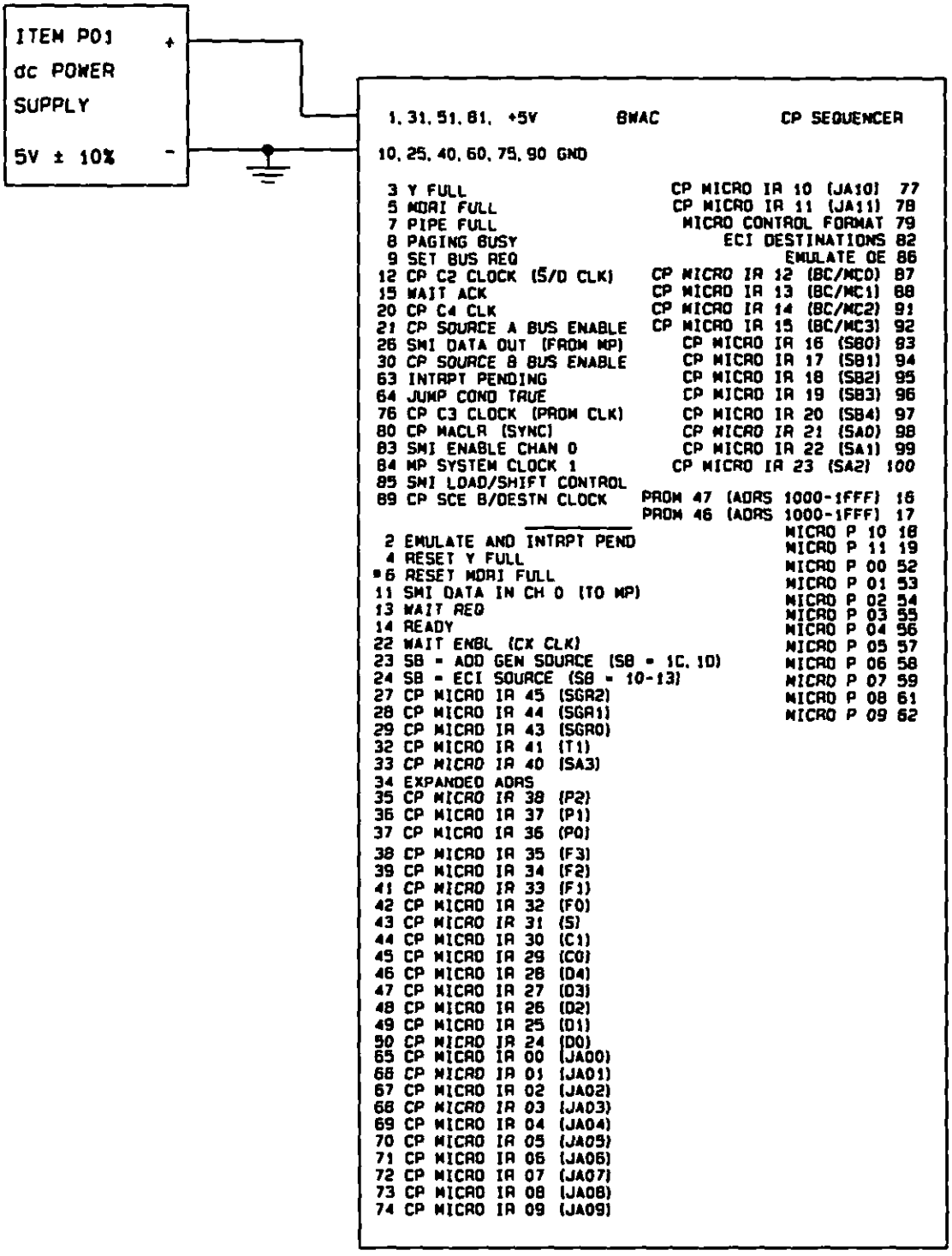
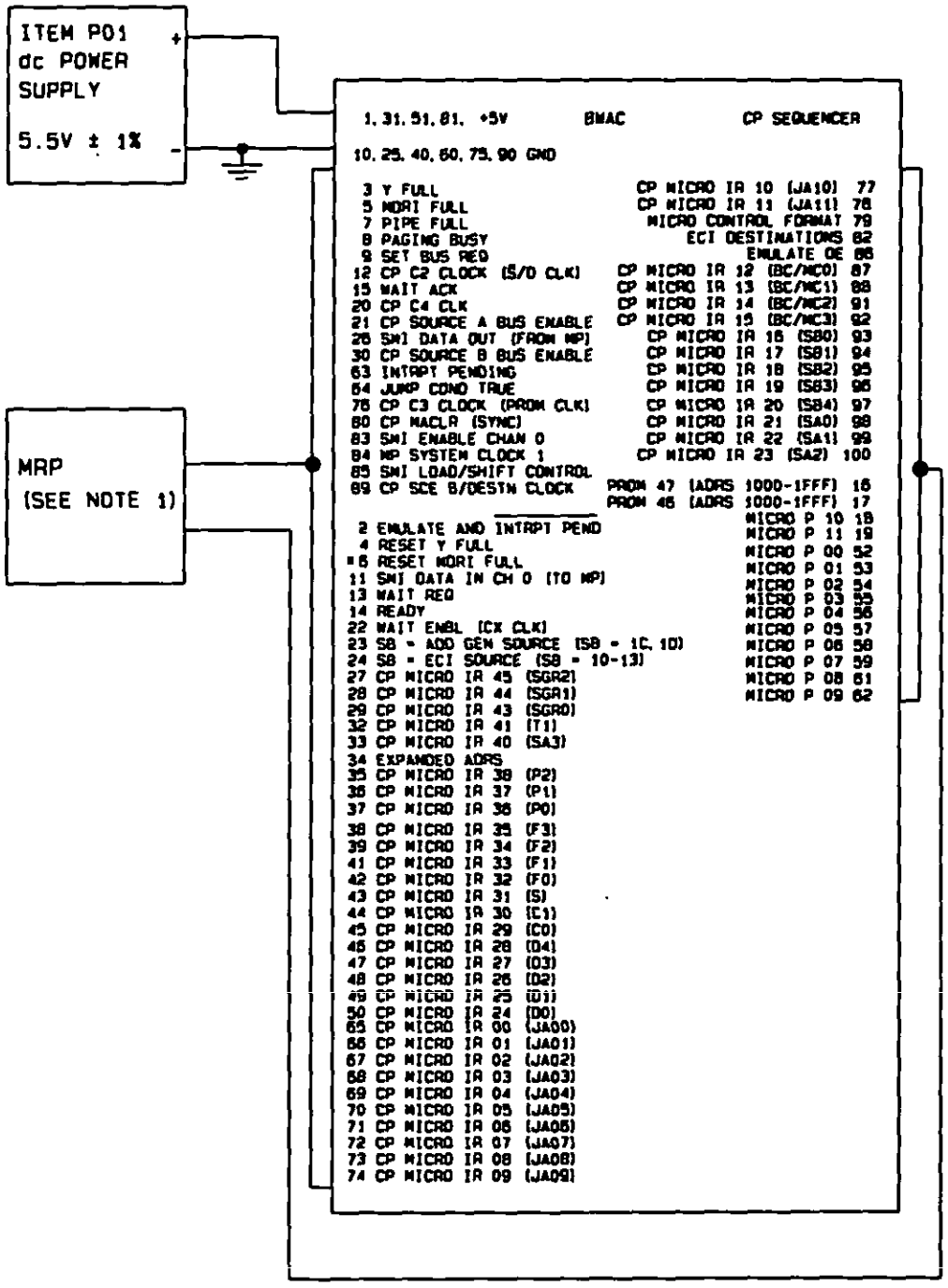


FIGURE 4. BWAC vibration test circuit.

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NOTE:

1. MRP or emulator capable of reproducing system backpanel characteristics.

FIGURE 5. BWAC life test circuit.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Definitions. Negative (-) current is current directed out of the module pin and positive (+) current is current directed into the module pin. Abbreviations used in this specification are defined as follows:

(a) Ci/o = input/output capacitance.
(b) Cin = input capacitance.
(c) Cout = output capacitance.
(d) Ibd = input breakdown current.
(e) Icc = power supply current.
(f) Iih = input high level current.
(g) Iil = input low level current.
(h) Iin = input forcing current.
(i) Ioh = output high level current.
(j) Iol = output low level current.
(k) Ios = output short circuit current.
(l) NA = not applicable.
(m) PSFC = power supply filter capacitance.
(n) PSTA = power supply transient amplitude.
(o) Vic = input clamp diode voltage.
(p) Vih = input high level voltage.
(q) Vil = input low level voltage.
(r) Vin = input forcing voltage.
(s) Voh = output high level voltage.
(t) Vol = output low level voltage.
(u) Vout = output forcing voltage.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements in table I, the module is considered to have completed its useful life (see 3.9).

6.3 Module description. The following electrical parameters reflect the -55°C to 85°C end-of-life parameters in table I. Operating power supply voltage range: 4.5 V dc to 5.5 V dc. Module power dissipation: 13.99 W (max); 10.81 W (typical).

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6.3.1 Input parameters.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
Y-FULL	-1.2	0.4	2.4	-2.0	0.05	1.0	50
MDRI-FULL	-1.2	0.4	2.4	-2.0	0.05	1.0	50
PIPE-FULL	-1.2	0.4	2.4	-0.6	0.02	0.1	50
PAGING-BUSY	-1.2	0.4	2.4	-0.6	0.02	0.1	50
SET-BUS-REQ	-1.2	0.4	2.4	-0.6	0.02	0.1	50
SD-CLK	-1.2	0.4	2.4	-5.8	0.16	2.3	100
WAIT-ACK	-1.2	0.4	2.4	-2.0	0.05	1.0	50
CP-C4-CLK	-1.2	0.4	2.4	-0.6	0.02	0.1	50
CPSRC-A-B-EN	-1.2	0.4	2.4	-2.0	0.05	1.0	50
SMI-DTA-OUT	-1.5	0.4	2.4	-0.4	0.02	0.1	50
CPSCE-B-BS-E	-1.2	0.4	2.4	-0.6	0.02	0.1	50
INTRPT-PNDNG	-1.2	0.4	2.4	-2.4	0.07	1.1	75
JMP-CND-TRU	NA	0.4	2.4	-3.11	-0.52	2.12	175
CP-C3-CLK	-1.2	0.4	2.4	-2.6	0.07	1.1	100

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
CP-MACLR	-1.2	0.4	2.4	-0.6	0.02	0.1	50
SMI-EN-CH-0	NA	0.4	2.4	-5.75	-0.49	2.12	125
MP-SYS-CLK	-1.2	0.4	2.4	-4.0	0.1	2.0	75
SMI-LD-CTRL	-1.2	0.4	2.4	-2.0	0.05	1.0	75
CP-SCE-B-CLK	-1.2	0.4	2.4	-6.0	0.15	3.0	100
PROM-47	NA	0.4	2.4	-0.8	0.24	NA	NA
PROM-46	NA	0.4	2.4	-0.8	0.24	NA	NA
MCRO-P-10	NA	0.4	2.4	-3.65	0.21	NA	321
MCRO-P-11	NA	0.4	2.4	-3.67	0.23	NA	312
MICRO-P00	NA	0.4	2.4	-3.65	0.21	NA	364
MICRO-P01	NA	0.4	2.4	-3.65	0.21	NA	200
MICRO-P02	NA	0.4	2.4	-3.65	0.21	NA	273
MICRO-P03	NA	0.4	2.4	-3.65	0.21	NA	263
MICRO-P04	NA	0.4	2.4	-3.65	0.21	NA	262

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
MICRO-P05	NA	0.4	2.4	-3.65	0.21	NA	254
MICRO-P06	NA	0.4	2.4	-3.65	0.21	NA	261
MICRO-P07	NA	0.4	2.4	-3.65	0.21	NA	283
MICRO-P08	NA	0.4	2.4	-3.65	0.21	NA	279
MICRO-P09	NA	0.4	2.4	-3.65	0.21	NA	328

6.3.2 Output parameters.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Ios max (mA)	Cout max (pF)
* EM-&-INT-PND	2.4	0.5	-0.16	1.6	-40.0	NA
RESET-Y-FULL	2.4	0.5	-0.4	6.0	-60.0	NA
RST-MDRI-FUL	2.4	0.5	-0.4	6.0	-60.0	NA
SMI-D-IN-0	2.4	0.5	-0.16	1.6	-15.0	NA
WAIT-REQ	2.4	0.5	-0.4	6.0	-60.0	NA
READY	2.4	0.5	-0.4	6.0	-60.0	NA
WAIT-ENBLE	2.4	0.5	-0.4	6.0	-40.0	NA

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6.3.2 Output parameters - Continued.

	Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Ios max (mA)	Cout max (pF)
*	SBEQADDGENSC	2.4	0.5	-0.16	1.6	-60.0	NA
*	SB-EQ-ECI-SC	2.4	0.5	-0.16	1.6	-60.0	NA
	MCROIR-45	2.4	0.5	-0.16	1.6	-30.0	50
	MCROIR-44	2.4	0.5	-0.16	1.6	-30.0	50
	MCROIR-43	2.4	0.5	-0.16	1.6	-30.0	75
	MCROIR-41	2.4	0.5	-0.16	1.6	-30.0	75
	MCROIR-40	2.4	0.5	-0.16	1.6	-30.0	100
*	XPANDED-ADRS	2.4	0.5	-0.16	1.6	-40.0	204
	MCROIR-38	2.4	0.5	-0.16	1.6	-30.0	150
	MCROIR-37	2.4	0.5	-0.16	1.6	-30.0	170
	MCROIR-36	2.4	0.5	-0.16	1.6	-30.0	200
	MCROIR-35	2.4	0.5	-0.16	1.6	-30.0	50
	MCROIR-34	2.4	0.5	-0.16	1.6	-30.0	50
	MCROIR-33	2.4	0.5	-0.16	1.6	-30.0	50

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Ios max (mA)	Cout max (pF)
MCROIR-32	2.4	0.5	-0.16	1.6	-30.0	50
MCROIR-31	2.4	0.5	-0.16	1.6	-30.0	50
MCROIR-30	2.4	0.5	-0.16	1.6	-30.0	50
MCROIR-29	2.4	0.5	-0.16	1.6	-30.0	50
MCROIR-28	2.4	0.5	-0.16	1.6	-30.0	75
MCROIR-27	2.4	0.5	-0.16	1.6	-30.0	75
MCROIR-26	2.4	0.5	-0.16	1.6	-30.0	75
MCROIR-25	2.4	0.5	-0.16	1.6	-30.0	75
MCROIR-24	2.4	0.5	-0.16	1.6	-30.0	75
MCROIR-00	2.4	0.5	-0.16	1.6	-30.0	75
MCROIR-01	2.4	0.5	-0.16	1.6	-30.0	75
MCROIR-02	2.4	0.5	-0.16	1.6	-30.0	75
MCROIR-03	2.4	0.5	-0.16	1.6	-30.0	75
MCROIR-04	2.4	0.5	-0.16	1.6	-30.0	75

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Ios max (mA)	Cout max (pF)
MCROIR-05	2.4	0.5	-0.16	1.6	-30.0	75
MCROIR-06	2.4	0.5	-0.16	1.6	-30.0	75
MCROIR-07	2.4	0.5	-0.16	1.6	-30.0	75
MCROIR-08	2.4	0.5	-0.16	1.6	-30.0	75
MCROIR-09	2.4	0.5	-0.16	1.6	-30.0	75
MCROIR-10	2.4	0.5	-0.16	1.6	-30.0	75
MCROIR-11	2.4	0.5	-0.16	1.6	-30.0	75
MCRO-CNT-FMT	2.4	0.5	-0.16	1.6	-30.0	NA
ECI-DESTNS	2.4	0.5	-0.16	1.6	-30.0	NA
* EMULATE-OE	2.4	0.5	-0.16	1.6	-60.0	NA
MCROIR-12	2.4	0.5	-0.16	1.6	-30.0	75
MCROIR-13	2.4	0.5	-0.16	1.6	-30.0	50
MCROIR-14	2.4	0.5	-0.16	1.6	-30.0	75
MCROIR-15	2.4	0.5	-0.16	1.6	NA	75

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6.3.2 Output parameters - Continued.

	Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Ios max (mA)	Cout max (pF)
	MCROIR-16	2.4	0.5	-0.16	1.6	-30.0	150
	MCROIR-17	2.4	0.5	-0.16	1.6	-30.0	200
	MCROIR-18	2.4	0.5	-0.16	1.6	-30.0	75
	MCROIR-19	2.4	0.5	-0.16	1.6	-30.0	75
	MCROIR-20	2.4	0.5	-0.16	1.6	-30.0	75
	MCROIR-21	2.4	0.5	-0.16	1.6	-30.0	75
	MCROIR-22	2.4	0.5	-0.16	1.6	-30.0	75
	MCROIR-23	2.4	0.5	-0.16	1.6	-30.0	75
	PROM-47	2.4	0.5	-0.16	1.6	-12.0	NA
	PROM-46	2.4	0.5	-0.16	1.6	-12.0	NA
*	MCRO-P-10	2.4	0.5	-0.16	1.6	-30.0	321
	MCRO-P-11	2.4	0.5	-0.16	1.6	-30.0	312
*	MICRO-P00	2.4	0.5	-0.16	1.6	-30.0	364
*	MICRO-P01	2.4	0.5	-0.16	1.6	-30.0	200

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6.3.2 Output parameters - Continued.

	Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Ios max (mA)	Cout max (pF)
*	MICRO-P02	2.4	0.5	-0.16	1.6	-30.0	273
*	MICRO-P03	2.4	0.5	-0.16	1.6	-30.0	263
*	MICRO-P04	2.4	0.5	-0.16	1.6	-30.0	262
*	MICRO-P05	2.4	0.5	-0.16	1.6	-30.0	254
*	MICRO-P06	2.4	0.5	-0.16	1.6	-30.0	261
*	MICRO-P07	2.4	0.5	-0.16	1.6	-30.0	283
*	MICRO-P08	2.4	0.5	-0.16	1.6	-30.0	279
*	MICRO-P09	2.4	0.5	-0.16	1.6	-30.0	328

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6.3.3 Class I and II dynamic parameters.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1	tpd	CP-SCE-B-CLK to MCROIR-45	-	-	-	50	50	50	nS	12
		CP-SCE-B-CLK to MCROIR-44	-	-	-					
		CP-SCE-B-CLK to MCROIR-43	-	-	-					
		CP-SCE-B-CLK to MCROIR-41	-	-	-					
		CP-SCE-B-CLK to MCROIR-40	-	-	-					
		CP-SCE-B-CLK to MCROIR-38	-	-	-					
		CP-SCE-B-CLK to MCROIR-37	-	-	-					
		CP-SCE-B-CLK to MCROIR-36	-	-	-					
		CP-SCE-B-CLK to MCROIR-35	-	-	-					
		CP-SCE-B-CLK to MCROIR-34	-	-	-					
		CP-SCE-B-CLK to MCROIR-33	-	-	-					
		CP-SCE-B-CLK to MCROIR-32	-	-	-					
		CP-SCE-B-CLK to MCROIR-31	-	-	-					
1	tpd	CP-SCE-B-CLK to MCROIR-30	-	-	-	50	50	50	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1	tpd	CP-SCE-B-CLK to MCROIR-29	-	-	-	50	50	50	nS	12
		CP-SCE-B-CLK to MCROIR-28	-	-	-					
		CP-SCE-B-CLK to MCROIR-27	-	-	-					
		CP-SCE-B-CLK to MCROIR-26	-	-	-					
		CP-SCE-B-CLK to MCROIR-25	-	-	-					
		CP-SCE-B-CLK to MCROIR-24	-	-	-					
		CP-SCE-B-CLK to MCROIR-00	-	-	-					
		CP-SCE-B-CLK to MCROIR-01	-	-	-					
		CP-SCE-B-CLK to MCROIR-02	-	-	-					
		CP-SCE-B-CLK to MCROIR-03	-	-	-					
		CP-SCE-B-CLK to MCROIR-04	-	-	-					
		CP-SCE-B-CLK to MCROIR-05	-	-	-					
		CP-SCE-B-CLK to MCROIR-06	-	-	-					
1	tpd	CP-SCE-B-CLK to MCROIR-07	-	-	-	50	50	50	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1	tpd	CP-SCE-B-CLK to MCROIR-08	-	-	-	50	50	50	nS	12
		CP-SCE-B-CLK to MCROIR-09	-	-	-					
		CP-SCE-B-CLK to MCROIR-10	-	-	-					
		CP-SCE-B-CLK to MCROIR-11	-	-	-					
		CP-SCE-B-CLK to MCROIR-12	-	-	-					
		CP-SCE-B-CLK to MCROIR-13	-	-	-					
		CP-SCE-B-CLK to MCROIR-14	-	-	-					
		CP-SCE-B-CLK to MCROIR-15	-	-	-					
		CP-SCE-B-CLK to MCROIR-16	-	-	-					
		CP-SCE-B-CLK to MCROIR-17	-	-	-					
		CP-SCE-B-CLK to MCROIR-18	-	-	-					
		CP-SCE-B-CLK to MCROIR-19	-	-	-					
		CP-SCE-B-CLK to MCROIR-20	-	-	-					
1	tpd	CP-SCE-B-CLK to MCROIR-21	-	-	-	50	50	50	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

	Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
				-55°C	25°C	85°C	-55°C	25°C	85°C		
	1	tpd	CP-SCE-B-CLK to MCROIR-22	-	-	-	50	50	50	nS	12
	1		CP-SCE-B-CLK to MCROIR-23	-	-	-	50	50	50		
	2		CP-SCE-B-CLK to RST-MDRI-FULL	-	-	-	35	35	35		
	2		CP-SCE-B-CLK to RESET-Y-FULL	-	-	-	35	35	35		
*	3		CP-SCE-B-CLK to MCRO-CNT-FMT	-	-	-	50	50	50		
	4		CP-SCE-B-CLK to SBEQADDGENSC	-	-	-	65	65	65		
	4		CP-SCE-B-CLK to SB-EQ-ECI-SC	-	-	-	65	65	65		
	5		CP-SCE-B-CLK to EMULATE-OE	-	-	-	85	85	85		
			CP-SCE-B-CLK to WAIT-REQ	-	-	-					
	5		CP-SCE-B-CLK to READY	-	-	-	85	85	85		
	6		CP-SCE-B-CLK to EM-&-INT-PND	-	-	-	145	145	145		
	6	tpd	CP-SCE-B-CLK to MCRO-P-10	-	-	-	145	145	145	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
6	tpd	CP-SCE-B-CLK to MCRO-P-11	-	-	-	145	145	145	nS	12
		CP-SCE-B-CLK to MCRO-P00	-	-	-					
		CP-SCE-B-CLK to MCRO-P01	-	-	-					
		CP-SCE-B-CLK to MCRO-P02	-	-	-					
		CP-SCE-B-CLK to MCRO-P03	-	-	-					
		CP-SCE-B-CLK to MCRO-P04	-	-	-					
		CP-SCE-B-CLK to MCRO-P05	-	-	-					
		CP-SCE-B-CLK to MCRO-P06	-	-	-					
		CP-SCE-B-CLK to MCRO-P07	-	-	-					
		CP-SCE-B-CLK to MCRO-P08	-	-	-					
6		CP-SCE-B-CLK to MCRO-P09	-	-	-	145	145	145		
7		CP-SCE-B-CLK to PROM-47	-	-	-	225	225	225		
7		CP-SCE-B-CLK to PROM-46	-	-	-	225	225	225		
8	tpd	Y-FULL to WAIT-REQ	-	-	-	25	25	25	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
8	tpd	MDRI-FULL to WAIT-REQ	-	-	-	25	25	25	nS	12
		PIPE-FULL to WAIT-REQ	-	-	-					
		PAGING BUSY to WAIT-REQ	-	-	-					
		SET-BUS-REQ to WAIT-REQ	-	-	-					
8		SD-CLK to WAIT-REQ	-	-	-					
9		WAIT-ACK to READY	-	-	-					
10		SD-CLK to WAIT-ENABLE	-	-	-					
11		CP-C4-CLK to WAIT-ENABLE	-	-	-	25	25	25		
12	tpd	MP-SYS-CLK to SMI-D-IN-0	-	-	-	60	60	60		
13	tsu	JMP-CND-TRU to CP-SCE-B-CLK	125	125	125	-	-	-		
14	th	CP-SCE-B-CLK to JMP-CND-TRU	10	10	10	-	-	-		
15	tsu	INTRPT-PNDNG to CP-SCE-B-CLK	125	125	125	-	-	-	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
16	th	CP-SCE-B-CLK to INTRPT-PNDNG	10	10	10	-	-	-	nS	12
17	tsu	MCRO-P-10 to CP-SCE-B-CLK	85	85	85	-	-	-		
		MCRO-P-11 to CP-SCE-B-CLK				-	-	-		
		MICRO-P00 to CP-SCE-B CLK				-	-	-		
		MICRO-P01 to CP-SCE-B-CLK				-	-	-		
		MICRO-P02 to CP-SCE-B-CLK				-	-	-		
		MICRO-P03 to CP-SCE-B-CLK				-	-	-		
		MICRO-P04 to CP-SCE-B-CLK				-	-	-		
		MICRO-P05 to CP-SCE-B-CLK				-	-	-		
17	tsu	MICRO-P06 to CP-SCE-B-CLK	85	85	85	-	-	-	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
17	tsu	MICRO-P07 to CP-SCE-B-CLK	85	85	85	-	-	-	nS	12
		MICRO-P08 to CP-SCE-B-CLK				-	-	-		
17	tsu	MICRO-P09 to CP-SCE-B-CLK	85	85	85	-	-	-		
18	th	CP-SCE-B-CLK to MCRO-P-10	10	10	10	-	-	-		
		CP-SCE-B-CLK to MCRO-P-11				-	-	-		
		CP-SCE-B-CLK to MICRO-P00				-	-	-		
		CP-SCE-B-CLK to MICRO-P01				-	-	-		
		CP-SCE-B-CLK to MICRO-P02				-	-	-		
		CP-SCE-B-CLK to MICRO-P03				-	-	-		
		CP-SCE-B-CLK to MICRO-P04				-	-	-		
		CP-SCE-B-CLK to MICRO-P05				-	-	-		
		CP-SCE-B-CLK to MICRO-P06				-	-	-		
18	th	CP-SCE-B-CLK to MICRO-P07	10	10	10	-	-	-	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

	Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
				-55°C	25°C	85°C	-55°C	25°C	85°C		
	18	th	CP-SCE-B-CLK to MICRO-P08	10	10	10	-	-	-	nS	12
	18	th	CP-SCE-B-CLK to MICRO-P09	10	10	10	-	-	-		
	19	tsu	SMI-DTA-OUT to CP-SCE-B-CLK	25	25	25	-	-	-		
	20	th	CP-SCE-B-CLK to SMI-DTA-OUT	30	30	30	-	-	-		
*	21	tsu	SMI-EN-CH-0 to CP-SCE-B-CLK	153	153	153	-	-	-		
*	22	th	CP-SCE-B-CLK to SMI-EN-CH-0	30	30	30	-	-	-	nS	12

6.4 Module function. The Central Processor (CP) sequencer (BWAC) provides the micromemory and the microcontrol functions for the CP. Micromemory contains 4096 48-bit words of permanent memory that is used to store the CP microprogram. The 48-bit microinstruction register holds the microinstruction read from micromemory. The microprogram sequencer determines the micromemory addressing sequence. The sequencer addresses consecutive locations except when an Emulation Control Word (ECW) is processed, an interrupt occurs (the interrupt pointer address is formed in the jump address buffer), or a branch microinstruction is executed resulting in a jump. The 72-bit Serial Maintenance Interface (SMI) register is used by the maintenance processor (BWAH) for Built In Test (BIT) fault isolation and for control and display functions. The register contains the microinstruction, micromemory address and miscellaneous control bits. The register is loaded or read serially by the maintenance processor.

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6.4.1 CP sequencer. The CP sequencer module contains the microprogram sequencer, micromemory, microinstruction register, SMI register, and associated logic circuits.

6.4.1.1 Microprogram sequencer. The microprogram sequencer provides the 12-bit address to fetch microinstructions from micromemory. The sequencer contains an address register, a microprogram counter register, and a five word last-in, first-out stack. The sequencer can select an address from one of these three sources or from an external source (direct input). The source is determined by the instruction bits I3-I0 (P3 - P0 from microinstruction). The microprogram counter register contains the last address plus one and is used to fetch consecutive microinstructions. The five-word stack is used to execute nested subroutines. The direct input (from the Jump Memory Address Pointer (JMAP) register or jump address buffer) is used to jump to a new location in the microprogram. The address register is not used. Timing is provided by the Source/Destination (S/D) clock.

6.4.1.2 Micromemory. Micromemory contains the unalterable CP microprogram. It is a Programmable Read Only Memory (PROM) memory containing 4096 48-bit microinstructions.

6.4.1.3 Microinstruction register. The 48-bit microinstruction register holds the microinstruction read from micromemory or the SMI register.

6.4.1.4 SMI register. The 72-bit SMI register holds the CP sequencer SMI word. The SMI word contains the current microinstruction, micromemory address, and miscellaneous control signals. The SMI register is parallel loaded on each S/D clock pulse if SMI is not enabled. When the maintenance processor enables SMI, the SMI register is right shifted one bit position on each maintenance processor clock pulse. The data is transferred to/from the maintenance processor via the serial maintenance interface on the BIT Electronics (CWH) module.

6.4.1.5 JMAP register. The 12-bit JMAP register holds the ECW pointer address or the interrupt pointer address. The register is enabled when the microinstruction P-field equals 2 (emulate) to form the ECW pointer address. If an interrupt is pending when the microinstruction P-field equals 2 and bit 41 equals zero (emulate start), the interrupt pointer address is formed. The base address 400 is modified by the interrupt pointer (from emulation control 2 module) to jump to the appropriate interrupt routine.

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6.4.1.6 Jump address buffer. The 12-bit jump address buffer holds the jump address from a format 1 microinstruction. The buffer is enabled when the microinstruction P-field is not equal to 2 or 6.

6.4.1.7 Source B (SB) decoder. The SB decoder translates the microinstruction SB field to transfer the selected source to the SB/destination bus.

6.4.1.8 Clock stop control. Clock stop control stops the clock generator (ALU module) when a macroinstruction or operand has not been received from main memory. The clock stop control restarts the clock when the condition is removed.

6.4.1.9 Microcontrol format decoder. The decoder translates the microinstruction P and SA fields to determine if it is a microcontrol format instruction (P = 0, 2, 8, or E, and SA = F).

6.4.2 Description of I/O pins.

Signal name	Pin number	I/O	Description
Y-FULL	3	I	Y FULL
MDRI-FULL	5	I	MDRI FULL
PIPE-FULL	7	I	PIPE FULL
PAGING-BUSY	8	I	PAGING BUSY
SET-BUS-REQ	9	I	SET BUS REQ
SD-CLK	12	I	CP C2 CLOCK (S/D CLK)
WAIT-ACK	15	I	WAIT ACK
CP-C4-CLK	20	I	CP C4 CLK
CPSRC-A-B-EN	21	I	CP SOURCE A BUS ENABLE
SMI-DTA-OUT	26	I	SMI DATA OUT (FROM MP)
CPSCE-B-BS-E	30	I	CP SOURCE B BUS ENABLE

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6.4.2 Description of I/O pins.

Signal name	Pin number	I/O	Description
INTRPT-PNDNG	63	I	INTRPT PENDING
JMP-CND-TRU	64	I	JUMP COND TRUE
CP-C3-CLK	76	I	CP C3 CLOCK (PROM CLK)
CP-MACLR	80	I	CP MACLR (SYNC)
SMI-EN-CH-0	83	I	SMI ENABLE CH 0
MP-SYS-CLK	84	I	MP SYSTEM CLOCK 1
SMI-LD-CTRL	85	I	SMI LOAD/SHIFT CONTROL
CP-SCE-B-CLK	89	I	CP SCE B/DESTN CLOCK
EM-&-INT-PND	2	O	EMULATE AND $\overline{\text{INTERRUPT PENDING}}$
RESET-Y-FULL	4	O	RESET Y FULL
RST-MDRI-FUL	6	O	RESET MDRI FULL
SMI-D-IN-0	11	O	SMI DATA IN CH 0 (TO MP)
WAIT-REQ	13	O	WAIT REQ
READY	14	O	READY
WAIT-ENABLE	22	O	WAIT ENABLE (CX CLK)
SBEQADDGENSC	23	O	SB = ADD GEN SOURCE (SB=1C,1D)
SB-EQ-ECI-SC	24	O	SB = ECI SOURCE (SB=10-13)
MCROIR-45	27	O	CP MICRO IR 45 (SGR2)
MCROIR-44	28	O	CP MICRO IR 44 (SGR1)
MCROIR-43	29	O	CP MICRO IR 43 (SGR0)
MCROIR-41	32	O	CP MICRO IR 41 (T1)

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6.4.2 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
MCROIR-40	33	O	CP MICRO IR 40 (SA3)
XPANDED-ADRS	34	O	EXPANDED ADRS
MCROIR-38	35	O	CP MICRO IR 38 (P2)
MCROIR-37	36	O	CP MICRO IR 37 (P1)
MCROIR-36	37	O	CP MICRO IR 36 (P0)
MCROIR-35	38	O	CP MICRO IR 35 (F3)
MCROIR-34	39	O	CP MICRO IR 34 (F2)
MCROIR-33	41	O	CP MICRO IR 33 (F1)
MCROIR-32	42	O	CP MICRO IR 32 (F0)
MCROIR-31	43	O	CP MICRO IR 31 (S)
MCROIR-30	44	O	CP MICRO IR 30 (C1)
MCROIR-29	45	O	CP MICRO IR 29 (C0)
MCROIR-28	46	O	CP MICRO IR 28 (D4)
MCROIR-27	47	O	CP MICRO IR 27 (D3)
MCROIR-26	48	O	CP MICRO IR 26 (D2)
MCROIR-25	49	O	CP MICRO IR 25 (D1)
MCROIR-24	50	O	CP MICRO IR 24 (D0)
MCROIR-00	65	O	CP MICRO IR 00 (JA00)
MCROIR-01	66	O	CP MICRO IR 01 (JA01)
MCROIR-02	67	O	CP MICRO IR 02 (JA02)
MCROIR-03	68	O	CP MICRO IR 03 (JA03)

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6.4.2 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
MCROIR-04	69	O	CP MICRO IR 04 (JA04)
MCROIR-05	70	O	CP MICRO IR 05 (JA05)
MCROIR-06	71	O	CP MICRO IR 06 (JA06)
MCROIR-07	72	O	CP MICRO IR 07 (JA07)
MCROIR-08	73	O	CP MICRO IR 08 (JA08)
MCROIR-09	74	O	CP MICRO IR 09 (JA09)
MCROIR-10	77	O	CP MICRO IR 10 (JA10)
MCROIR-11	78	O	CP MICRO IR 11 (JA11)
MCRO-CNT-FMT	79	O	MICRO CONTROL FORMAT
ECI-DESTNS	82	O	ECI DESTINATIONS
EMULATE-OE	86	O	EMULATE OE
MCROIR-12	87	O	CP MICRO IR 12 (BC/MC0)
MCROIR-13	88	O	CP MICRO IR 13 (BC/MC1)
MCROIR-14	91	O	CP MICRO IR 14 (BC/MC2)
MCROIR-15	92	O	CP MICRO IR 15 (BC/MC3)
MCROIR-16	93	O	CP MICRO IR 16 (SB0)
MCROIR-17	94	O	CP MICRO IR 17 (SB1)
MCROIR-18	95	O	CP MICRO IR 18 (SB2)
MCROIR-19	96	O	CP MICRO IR 19 (SB3)
MCROIR-20	97	O	CP MICRO IR 20 (SB4)
MCROIR-21	98	O	CP MICRO IR 21 (SA0)

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6.4.2 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
MCROIR-22	99	O	CP MICRO IR 22 (SA1)
MCROIR-23	100	O	CP MICRO IR 23 (SA2)
PROM-47	16	I/O	PROM 47 (ADRS 1000-1FFF)
PROM-46	17	I/O	PROM 46 (ADRS 1000-1FFF)
MCRO-P-10	18	I/O	MICRO P 10
MCRO-P-11	19	I/O	MICRO P 11
MICRO-P00	52	I/O	MICRO P 00
MICRO-P01	53	I/O	MICRO P 01
MICRO-P02	54	I/O	MICRO P 02
MICRO-P03	55	I/O	MICRO P 03
MICRO-P04	56	I/O	MICRO P 04
MICRO-P05	57	I/O	MICRO P 05
MICRO-P06	58	I/O	MICRO P 06
MICRO-P07	59	I/O	MICRO P 07
MICRO-P08	61	I/O	MICRO P 08
MICRO-P09	62	I/O	MICRO P 09

6.5 Functional test vectors. Functional test vectors are not printed herein because of their extreme length; and, even when presented as a hard copy, they require conversion to on-line format. When the entire sequence of vector groups is needed for driving devices as part of inspection, a test program output should be generated in a form compatible with test system architecture (magnetic tape format usually preferred). For reference purposes, the Naval Surface Warfare

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Center, Crane Division (Code 603) maintains a limited quantity of dated, hard copy printouts of the functional test vectors and these may be obtained upon request.

6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

*Custodians:
Army - ER
Navy - NW
Air force - 85

Review activities:
Army - AT
Navy - AS, MC
Air force - 13,19
DLA - ES

Preparing activity:
Navy - NW

(Project 5963-0281)

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TABLE II. Isolation test.

	Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
*	A	Isolation, frame	C	Frame	10	INF	MOhm	1-8

NOTES:

1. Isolation test tolerance table.

Condition	Value	Accuracy	Test condition
Megohm bridge voltage	100 V	± 10 percent	A
Isolation measurement	NA	± 500 KOhm	A

2. The ohmmeter used shall be item R06.

3. In the set-up conditions in table II, 'A' shall correspond to the positive terminal of the megohm bridge, 'B' shall correspond to the negative terminal and 'C' shall correspond to all used pins shorted together.

4. Warning: When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "DISCHARGE" before connecting or disconnecting the test leads.

5. When testing for frame isolation, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal of the megohm bridge.

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NOTES: - Continued.

6. Caution: Set megohm bridge to limit current to 10 μA before applying voltage to the module.

7. The test circuit shall be as indicated on figure 6.

8. Abbreviations used are defined in 6.1.

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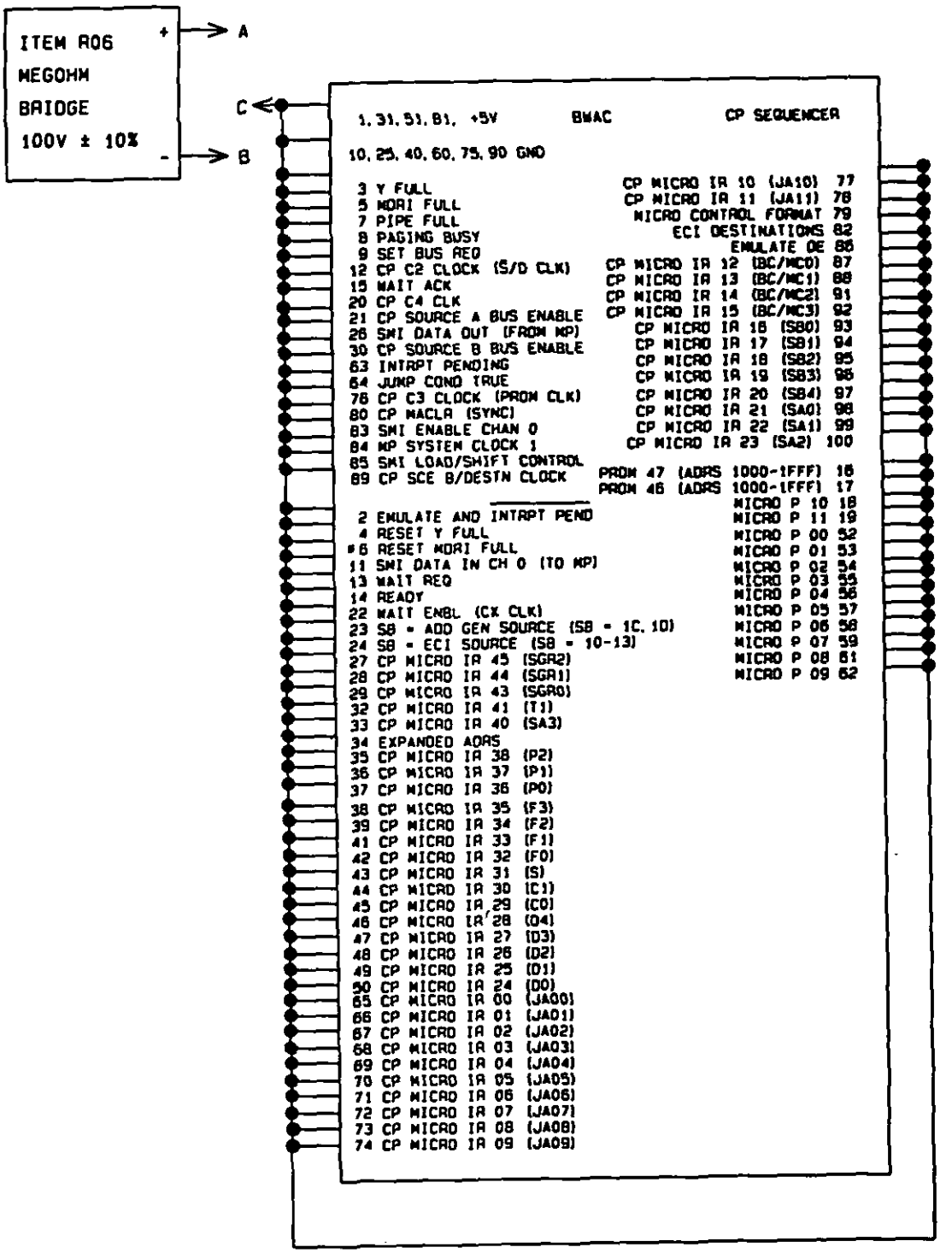


FIGURE 6. Isolation test circuit for table II.

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TABLE III. Continuity test.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
B1	Continuity, ground	10	25	0	1	Ohm	1-5
B2	Continuity, ground	10	40				
B3	Continuity, ground	60	75				
B4	Continuity, ground	60	90				
B5	Continuity, +5V	1	31				
B6	Continuity, +5V	51	81	0	1	Ohm	1-5

NOTES:

1. Continuity test tolerance table.

Condition	Value	Accuracy	Test condition
Continuity measurement	NA	± 0.05 Ohm	B1-B6

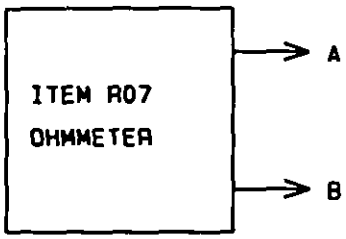
2. The ohmmeter used shall be item R07.

3. In the set-up conditions in table III, 'A' shall correspond to the positive terminal of the ohmmeter, and 'B' shall correspond to the negative terminal.

4. The test circuit shall be as indicated on figure 7.

5. Abbreviations used are defined in 6.1.

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1, 31, 51, 81, +5V		BWAC	CP SEQUENCER
10, 25, 40, 60, 75, 90 GND			
3 Y FULL		CP MICRO IR 10 (JA10)	77
5 MDRI FULL		CP MICRO IR 11 (JA11)	78
7 PIPE FULL		MICRO CONTROL FORMAT	79
8 PAGING BUSY		ECI DESTINATIONS	82
9 SET BUS REQ		EMULATE OE	86
12 CP C2 CLOCK (S/D CLK)		CP MICRO IR 12 (BC/MC0)	87
15 WAIT ACK		CP MICRO IR 13 (BC/MC1)	88
20 CP C4 CLK		CP MICRO IR 14 (BC/MC2)	91
21 CP SOURCE A BUS ENABLE		CP MICRO IR 15 (BC/MC3)	92
26 SMI DATA OUT (FROM MP)		CP MICRO IR 16 (SB0)	93
30 CP SOURCE B BUS ENABLE		CP MICRO IR 17 (SB1)	94
63 INTRPT PENDING		CP MICRO IR 18 (SB2)	95
64 JUMP COND TRUE		CP MICRO IR 19 (SB3)	96
76 CP C3 CLOCK (PROM CLK)		CP MICRO IR 20 (SB4)	97
80 CP MACLR (SYNC)		CP MICRO IR 21 (SA0)	98
83 SMI ENABLE CHAN 0		CP MICRO IR 22 (SA1)	99
84 MP SYSTEM CLOCK 1		CP MICRO IR 23 (SA2)	100
85 SMI LOAD/SHIFT CONTROL			
89 CP SCE 0/DESTN CLOCK		PROM 47 (ADRS 1000-1FFF)	16
		PROM 46 (ADRS 1000-1FFF)	17
2 EMULATE AND INTRPT PEND		MICRO P 10	18
4 RESET Y FULL		MICRO P 11	19
*6 RESET MDRI FULL		MICRO P 00	52
11 SMI DATA IN CH 0 (TO MP)		MICRO P 01	53
13 WAIT REQ		MICRO P 02	54
14 READY		MICRO P 03	55
22 WAIT ENBL (CX CLK)		MICRO P 04	56
23 SB = ADD GEN SOURCE (SB = 1C, 1D)		MICRO P 05	57
24 SB = ECI SOURCE (SB = 10-13)		MICRO P 06	58
27 CP MICRO IR 45 (SGR2)		MICRO P 07	59
28 CP MICRO IR 44 (SGR1)		MICRO P 08	61
29 CP MICRO IR 43 (SGR0)		MICRO P 09	62
32 CP MICRO IR 41 (T1)			
33 CP MICRO IR 40 (SA3)			
34 EXPANDED ADRS			
35 CP MICRO IR 38 (P2)			
36 CP MICRO IR 37 (P1)			
37 CP MICRO IR 36 (P0)			
38 CP MICRO IR 35 (F3)			
39 CP MICRO IR 34 (F2)			
41 CP MICRO IR 33 (F1)			
42 CP MICRO IR 32 (F0)			
43 CP MICRO IR 31 (S)			
44 CP MICRO IR 30 (C1)			
45 CP MICRO IR 29 (C0)			
46 CP MICRO IR 28 (D4)			
47 CP MICRO IR 27 (D3)			
48 CP MICRO IR 26 (D2)			
49 CP MICRO IR 25 (D1)			
50 CP MICRO IR 24 (D0)			
65 CP MICRO IR 00 (JA00)			
66 CP MICRO IR 01 (JA01)			
67 CP MICRO IR 02 (JA02)			
68 CP MICRO IR 03 (JA03)			
69 CP MICRO IR 04 (JA04)			
70 CP MICRO IR 05 (JA05)			
71 CP MICRO IR 06 (JA06)			
72 CP MICRO IR 07 (JA07)			
73 CP MICRO IR 08 (JA08)			
74 CP MICRO IR 09 (JA09)			

FIGURE 7. Continuity test circuit for table III.

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TABLE IV. Power supply stress test.

Test cond	Power supply	Connect Positive lead to	Connect Negative lead to	25 ± 5°C Initial limits	Notes
C	Vcc stress	1,31, 51,81	10,25, 40,60, 75,90	Module shall pass all remaining tests.	1-6

NOTES:

1. Power supply stress test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	7.0 V	+0 mV, -70 mV	C
Power supply current limit	4.58 A	NA	C

2. There shall be no power supply voltage overshoot on this test.

3. Connect the power supplies to the module as indicated in table IV.

4. When performing the power supply stress test, no measurements are taken. This condition shall be maintained for a minimum of 5 seconds.

5. The test circuit shall be as indicated on figure 8.

6. Abbreviations used are defined in 6.1.

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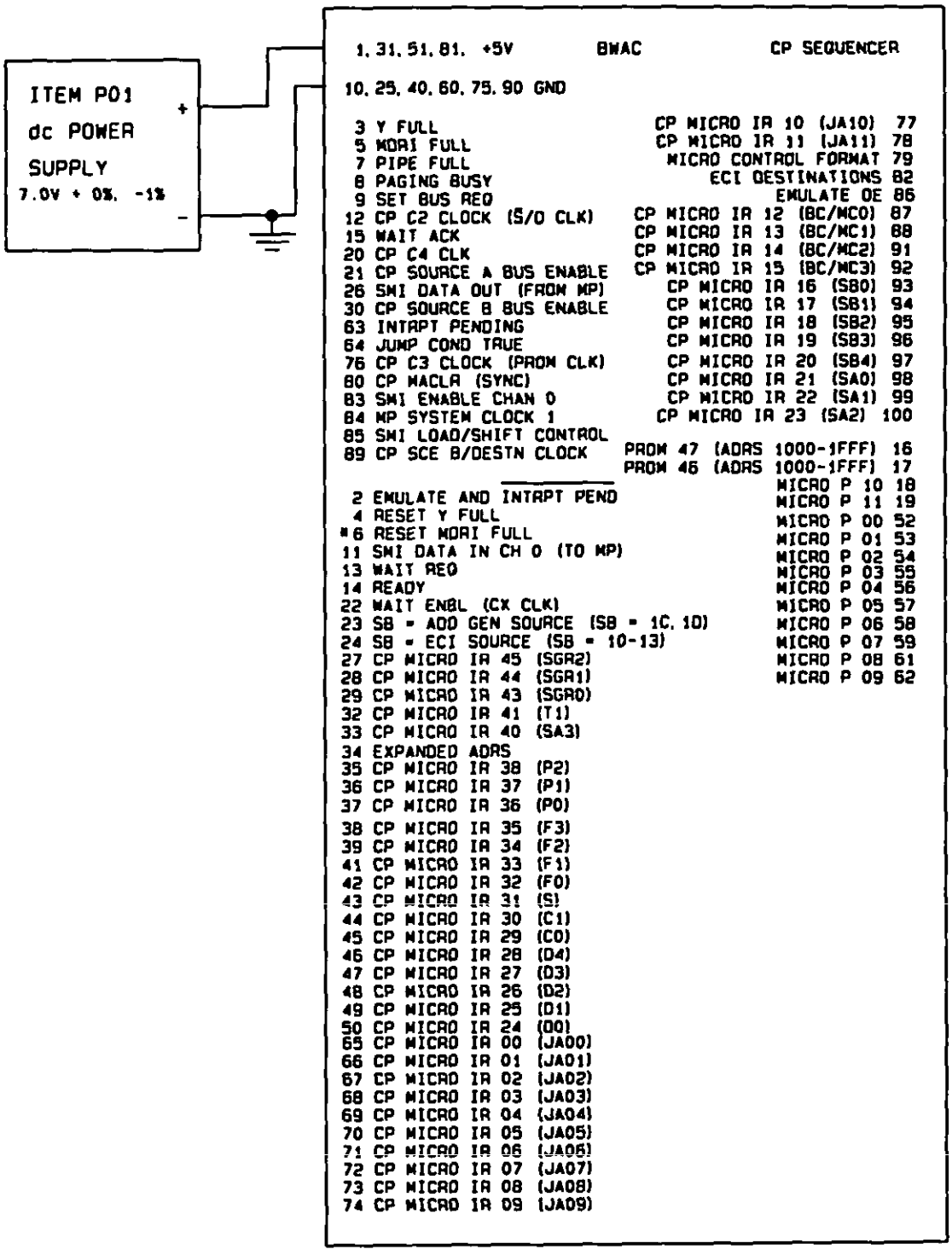


FIGURE 8. Vcc stress test circuit for table IV.

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TABLE V. Input breakdown current test.

	Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
	D1	Y-FULL	3	5.5	0.0	1.0	mA	1-3
	D2	MDRI-FULL	5			1.0		
	D3	PIPE-FULL	7			0.1		
	D4	PAGING-BUSY	8					
	D5	SET-BUS-REQ	9			0.1		
	D6	SD-CLK	12			2.3		
	D7	WAIT-ACK	15			1.0		
*	D8	CP-C4-CLK	20			0.1		
	D9	CPSRC-A-B-EN	21			1.0		
	D10	SMI-DTA-OUT	26			0.1		
	D11	CPSCE-B-BS-E	30			0.1		
	D12	INTRPT-PNDNG	63		0.0	1.1		
	D13	JMP-CND-TRU	64		-0.02	2.12		
	D14	CP-C3-CLOCK	76		0.0	1.1		
	D15	CP-MACLR	80		0.0	0.1		
	D16	SMI-EN-CH-0	83		-0.02	2.12		
	D17	MP-SYS-CLK	84		0.0	2.0		
	D18	SMI-LD-CTRL	85			1.0		
	D19	CP-SCE-B-CLK	89	5.5	0.0	3.0	mA	1-3

Source: https://assist.dla.mil -- Downloaded: 20260-20T18:24Z
Check the source to verify that this is the current version before use.

NOTES:

1. Input breakdown current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	D1-D19
Power supply current limit	3.82 A	NA	D1-D19
Forcing voltage, (Vin)	5.5 V	± 55 mV	D1-D19
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 D1-D19 D1-D19
Input current measurement	NA NA	± 50 µA ± 5 µA ± 115 µA ± 50 µA ± 5 µA ± 50 µA ± 5 µA ± 55 µA ± 107 µA ± 55 µA ± 5 µA ± 107 µA ± 100 µA ± 50 µA ± 150 µA	D1,D2 D3-D5 D6 D7 D8 D9 D10,D11 D12 D13 D14 D15 D16 D17 D18 D19

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Check the source to verify that this is the current version before use.

NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table V and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:24Z
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TABLE VI. Output short circuit current test.

	Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
	E1	EM-&-INT-PND	2	0.0	-100.0	-40.0	mA	1-4
	E2	RESET-Y-FULL	4		-150.0	-60.0		
	E3	RST-MDRI-FUL	6		-150.0	-60.0		
	E4	SMI-D-IN-0	11		-130.0	-15.0		
	E5	WAIT-REQ	13		-150.0	-60.0		
	E6	READY	14		-150.0	-60.0		
*	E7	PROM-47	16		-160.0	-12.0		
*	E8	PROM-47	16					
*	E9	PROM-46	17					
*	E10	PROM-46	17		-160.0	-12.0		
	E11	MCRO-P-10	18		-85.0	-30.0		
	E12	MCRO-P-11	19		-85.0	-30.0		
	E13	WAIT-ENABLE	22		-100.0	-40.0		
	E14	SBEQADDGENSC	23		-150.0	-60.0		
	E15	SB-EQ-ECI-SC	24		-150.0	-60.0		
	E16	MCROIR-45	27		-260.0	-30.0		
	E17	MCROIR-44	28					
	E18	MCROIR-43	29					
	E19	MCROIR-41	32					
	E20	MCROIR-40	33		-260.0	-30.0		
	E21	XPANDED-ADRS	34	0.0	-100.0	-40.0	mA	1-4

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TABLE VI. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
E22	MCROIR-38	35	0.0	-260.0	-30.0	mA	1-4
E23	MCROIR-37	36					
E24	MCROIR-36	37					
E25	MCROIR-35	38					
E26	MCROIR-34	39					
E27	MCROIR-33	41					
E28	MCROIR-32	42					
E29	MCROIR-31	43					
E30	MCROIR-30	44					
E31	MCROIR-29	45					
E32	MCROIR-28	46					
E33	MCROIR-27	47					
E34	MCROIR-26	48					
E35	MCROIR-25	49					
E36	MCROIR-24	50		-260.0			
E37	MICRO-P00	52		-85.0			
E38	MICRO-P01	53					
E39	MICRO-P02	54					
E40	MICRO-P03	55					
E41	MICRO-P04	56					
E42	MICRO-P05	57	0.0	-85.0	-30.0	mA	1-4

Source: https://assist.dla.mil -- Downloaded: 20260-20T18:24Z
Check the source to verify that this is the current version before use.

TABLE VI. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E43	MICRO-P06	58	0.0	-85.0	-30.0	mA	1-4
E44	MICRO-P07	59					
E45	MICRO-P08	61					
E46	MICRO-P09	62		-85.0			
E47	MCROIR-00	65		-260.0			
E48	MCROIR-01	66					
E49	MCROIR-02	67					
E50	MCROIR-03	68					
E51	MCROIR-04	69					
E52	MCROIR-05	70					
E53	MCROIR-06	71					
E54	MCROIR-07	72					
E55	MCROIR-08	73					
E56	MCROIR-09	74					
E57	MCROIR-10	77					
E58	MCROIR-11	78		-260.0			
E59	MCRO-CNT-FMT	79		-130.0			
E60	ECI-DESTNS	82		-130.0	-30.0		
E61	EMULATE-OE	86		-150.0	-60.0		
E62	MCROIR-12	87		-260.0	-30.0		
E63	MCROIR-13	88	0.0	-260.0	-30.0	mA	1-4

Source: https://assist.dla.mil -- Downloaded: 20260-20T18:24Z
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TABLE VI. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
E64	MCROIR-14	91	0.0	-260.0	-30.0	mA	1-4
E65	MCROIR-15	92					
E66	MCROIR-16	93					
E67	MCROIR-17	94					
E68	MCROIR-18	95					
E69	MCROIR-19	96					
E70	MCROIR-20	97					
E71	MCROIR-21	98					
E72	MCROIR-22	99					
E73	MCROIR-23	100	0.0	-260.0	-30.0	mA	1-4

NOTES:

1. Output short circuit current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	E1-E73
Power supply current limit	3.82 A	NA	E1-E73
Forcing voltage, (Vout)	0.0 V	± 50 mV	E1-E73

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Pattern voltage			
high, (VH)	3.0 V	± 100 mV	E1-E73
low, (VL)	0.0 V	± 100 mV	E1-E73
Output current measurement	NA	± 3.0 mA	E1
		± 4.5 mA	E2,E3
		± 5.75 mA	E4
		± 4.5 mA	E5,E6
		*± 7.4 mA	E7-E10
		± 2.75 mA	E11,E12
		± 3.0 mA	E13
		± 4.5 mA	E14,E15
		± 11.5 mA	E16-E20
		± 3.0 mA	E21
		± 11.5 mA	E22-E36
		± 2.75 mA	E37-E46
		± 11.5 mA	E47-E58
		± 5.0 mA	E59,E60
		± 4.5 mA	E61
	NA	± 11.5 mA	E62-E73

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

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NOTES: - Continued.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VI and the current then measured. The forcing voltage undershoot shall not exceed 100 mV.

3. When performing this test, only one output shall be tested at a time, and the duration of momentarily grounding each output shall not exceed 1 second.

4. Abbreviations used are defined in 6.1.

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TABLE VII. Input current, high level, test.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F1	Y-FULL	3	2.4	-2.0	0.05	mA	1-3
F2	MDRI-FULL	5		-2.0	0.05		
F3	PIPE-FULL	7		-0.6	0.02		
F4	PAGING-BUSY	8					
F5	SET-BUS-REQ	9		-0.6	0.02		
F6	SD-CLK	12		-5.8	0.16		
F7	WAIT-ACK	15		-2.0	0.05		
F8	PROM-47	16		-0.8	0.24		
F9	PROM-46	17		-0.8	0.24		
F10	MCRO-P-10	18		-3.65	0.21		
F11	MCRO-P-11	19		-3.67	0.23		
F12	CP-C4-CLK	20		-0.6	0.02		
F13	CPSRC-A-B-EN	21		-2.0	0.05		
F14	SMI-DTA-OUT	26		-0.4	0.02		
F15	CPSCE-B-BS-E	30		-0.6	0.02		
F16	MICRO-P00	52		-3.65	0.21		
F17	MICRO-P01	53					
F18	MICRO-P02	54					
F19	MICRO-P03	55					
F20	MICRO-P04	56					
F21	MICRO-P05	57	2.4	-3.65	0.21	mA	1-3

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Check the source to verify that this is the current version before use.

TABLE VII. Input current, high level, test - Continued.

	Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
	F22	MICRO-P06	58	2.4	-3.65	0.21	mA	1-3
*	F23	MICRO-P07	59					
*	F24	MICRO-P08	61					
	F25	MICRO-P09	62		-3.65	0.21		
	F26	INTRPT-PNDNG	63		-2.4	0.07		
	F27	JMP-CND-TRU	64		-3.11	-0.52		
	F28	CP-C3-CLK	76		-2.6	0.07		
	F29	CP-MACLR	80		-0.6	0.02		
	F30	SMI-EN-CH-0	83		-5.75	-0.49		
	F31	MP-SYS-CLK	84		-4.0	0.1		
	F32	SMI-LD-CTRL	85		-2.0	0.05		
	F33	CP-SCE-B-CLK	89	2.4	-6.0	0.15	mA	1-3

NOTES:

1. Input current, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	F1-F33
Power supply current limit	3.82 A	NA	F1-F33

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:24Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Forcing voltage, (Vih)	2.4 V	± 50 mV	F1-F33
Pattern voltage			
high, (VH)	3.0 V	± 100 mV	F1-F33
low, (VL)	0.0 V	± 100 mV	F1-F33
Input current measurement	NA	± 103 µA	F1,F2
		± 31 µA	F3-F5
		± 298 µA	F6
		± 103 µA	F7
		± 52 µA	F8,F9
		± 193 µA	F10
		± 195 µA	F11
		± 31 µA	F12
		± 103 µA	F13
		± 21 µA	F14
		± 31 µA	F15
		± 193 µA	F16-F25
		± 124 µA	F26
		± 130 µA	F27
		± 134 µA	F28
		± 31 µA	F29
		± 263 µA	F30
		± 205 µA	F31
		± 103 µA	F32
	NA	± 308 µA	F33

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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Check the source to verify that this is the current version before use.

NOTES: - Continued.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE VIII. Input current, low level, test.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G1	Y-FULL	3	0.4	-2.0	0.05	mA	1-3
G2	MDRI-FULL	5		-2.0	0.05		
G3	PIPE-FULL	7		-0.6	0.02		
G4	PAGING-BUSY	8					
G5	SET-BUS-REQ	9		-0.6	0.02		
G6	SD-CLK	12		-5.8	0.16		
G7	WAIT-ACK	15		-2.0	0.05		
G8	PROM-47	16		-0.8	0.24		
G9	PROM-46	17		-0.8	0.24		
G10	MCRO-P-10	18		-3.65	0.21		
G11	MCRO-P-11	19		-3.67	0.23		
G12	CP-C4-CLK	20		-0.6	0.02		
G13	CPSRC-A-B-EN	21		-2.0	0.05		
G14	SMI-DTA-OUT	26		-0.4	0.02		
G15	CPSCE-B-BS-E	30		-0.6	0.02		
G16	MICRO-P00	52		-3.65	0.21		
G17	MICRO-P01	53					
G18	MICRO-P02	54					
G19	MICRO-P03	55					
G20	MICRO-P04	56					
G21	MICRO-P05	57	0.4	-3.65	0.21	mA	1-3

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Check the source to verify that this is the current version before use.

TABLE VIII. Input current, low level, test - Continued.

	Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
	G22	MICRO-P06	58	0.4	-3.65	0.21	mA	1-3
*	G23	MICRO-P07	59					
*	G24	MICRO-P08	61					
	G25	MICRO-P09	62		-3.65	0.21		
	G26	INTRPT-PNDNG	63		-2.4	0.07		
	G27	JMP-CND-TRU	64		-3.11	-0.52		
	G28	CP-C3-CLK	76		-2.6	0.07		
	G29	CP-MACLR	80		-0.6	0.02		
	G30	SMI-EN-CH-0	83		-5.75	-0.49		
	G31	MP-SYS-CLK	84		-4.0	0.1		
	G32	SMI-LD-CTRL	85		-2.0	0.05		
	G33	CP-SCE-B-CLK	89	0.4	-6.0	0.15	mA	1-3

NOTES:

1. Input current, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	G1-G33
Power supply current limit	3.82 A	NA	G1-G33

Source: https://assist.dla.mil -- Downloaded: 20260-20T18:24Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Forcing voltage, (Vil)	0.4 V	± 50 mV	G1-G33
Pattern voltage high, (VH)	3.0 V	± 100 mV	G1-G33
low, (VL)	0.0 V	± 100 mV	G1-G33
Input current measurement	NA	± 103 µA	G1,G2
		± 31 µA	G3-G5
		± 298 µA	G6
		± 103 µA	G7
		± 52 µA	G8,G9
		± 193 µA	G10
		± 195 µA	G11
		± 40 µA	G12
		± 103 µA	G13
		± 21 µA	G14
		± 31 µA	G15
		± 193 µA	G16-G25
		± 124 µA	G26
		± 130 µA	G27
		± 134 µA	G28
		± 31 µA	G29
		± 263 µA	G30
		± 205 µA	G31
		± 103 µA	G32
	NA	± 308 µA	G33

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

Source: https://assist.dla.mil -- Downloaded: 20260-20T18:24Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VIII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:24Z
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TABLE IX. Output voltage, high level, test.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
H1	EM-&-INT-PND	2	-0.16	2.4	4.5	V	1-4
H2	RESET-Y-FULL	4	-0.4				
H3	RST-MDRI-FUL	6	-0.4				
H4	SMI-D-IN-0	11	-0.16				
H5	WAIT-REQ	13	-0.4				
H6	READY	14	-0.4				
H7	PROM-47	16	-0.16				
H8	PROM-47	16					
H9	PROM-46	17					
H10	PROM-46	17					
H11	MCRO-P-10	18					
H12	MCRO-P-11	19	-0.16				
H13	WAIT-ENABLE	22	-0.4				
H14	SBEQADDGENSC	23	-0.16				
H15	SB-EQ-ECI-SC	24					
H16	MCROIR-45	27					
H17	MCROIR-44	28					
H18	MCROIR-43	29					
H19	MCROIR-41	32					
H20	MCROIR-40	33					
H21	XPANDED-ADRS	34	-0.16	2.4	4.5	V	1-4

Source: https://assist.dla.mil -- Downloaded: 20260410-20T18:24Z
Check the source to verify that this is the current version before use.

TABLE IX. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
H22	MCROIR-38	35	-0.16	2.4	4.5	V	1-4
H23	MCROIR-37	36					
H24	MCROIR-36	37					
H25	MCROIR-35	38					
H26	MCROIR-34	39					
H27	MCROIR-33	41					
H28	MCROIR-32	42					
H29	MCROIR-31	43					
H30	MCROIR-30	44					
H31	MCROIR-29	45					
H32	MCROIR-28	46					
H33	MCROIR-27	47					
H34	MCROIR-26	48					
H35	MCROIR-25	49					
H36	MCROIR-24	50					
H37	MICRO-P00	52					
H38	MICRO-P01	53					
H39	MICRO-P02	54					
H40	MICRO-P03	55					
H41	MICRO-P04	56					
H42	MICRO-P05	57	-0.16	2.4	4.5	V	1-4

Source: https://assist.dla.mil -- Downloaded: 202610-20T18:24Z
Check the source to verify that this is the current version before use.

TABLE IX. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
H43	MICRO-P06	58	-0.16	2.4	4.5	V	1-4
H44	MICRO-P07	59					
H45	MICRO-P08	61					
H46	MICRO-P09	62					
H47	MCROIR-00	65					
H48	MCROIR-01	66					
H49	MCROIR-02	67					
H50	MCROIR-03	68					
H51	MCROIR-04	69					
H52	MCROIR-05	70					
H53	MCROIR-06	71					
H54	MCROIR-07	72					
H55	MCROIR-08	73					
H56	MCROIR-09	74					
H57	MCROIR-10	77					
H58	MCROIR-11	78					
H59	MCRO-CNT-FMT	79					
H60	ECI-DESTNS	82					
H61	EMULATE-OE	86					
H62	MCROIR-12	87					
H63	MCROIR-13	88	-0.16	2.4	4.5	V	1-4

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Check the source to verify that this is the current version before use.

TABLE IX. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
H64	MCROIR-14	91	-0.16	2.4	4.5	V	1-4
H65	MCROIR-15	92					
H66	MCROIR-16	93					
H67	MCROIR-17	94					
H68	MCROIR-18	95					
H69	MCROIR-19	96					
H70	MCROIR-20	97					
H71	MCROIR-21	98					
H72	MCROIR-22	99					
H73	MCROIR-23	100	-0.16	2.4	4.5	V	1-4

NOTES:

1. Output voltage, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	H1-H73
Power supply current limit	3.82 A	NA	H1-H73

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Forcing current, (Ioh)	-0.16 mA	± 16 µA	H1
	-0.4 mA	± 40 µA	H2,H3
	-0.16 mA	± 16 µA	H4
	-0.4 mA	± 40 µA	H5,H6
	-0.16 mA	± 16 µA	H7-H12
	-0.4 mA	± 40 µA	H13
	-0.16 mA	± 16 µA	H14-H73
Pattern voltage high, (VH)	3.0 V	± 100 mV	H1-H73
low, (VL)	0.0 V	± 100 mV	H1-H73
Output voltage measurement	NA	± 105 mV	H1-H73

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

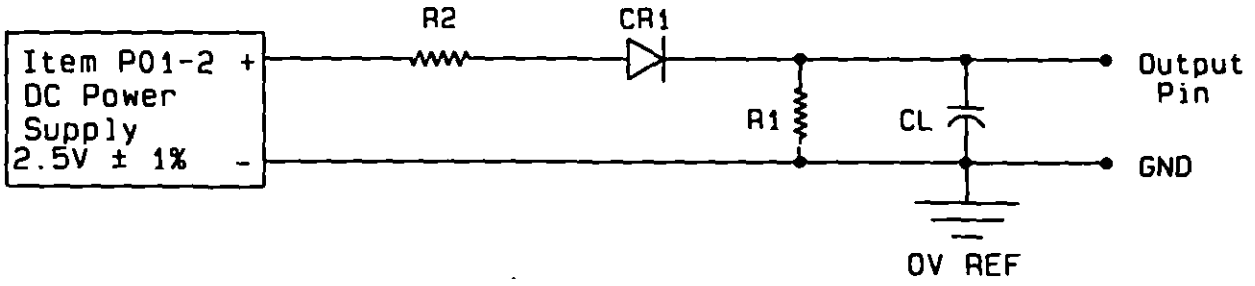
(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table IX and the voltage then measured.

3. The load shall be as indicated on figure 9.

4. Abbreviations used are defined in 6.1.

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NOTES:

1. CR1 = 1N4148 or equivalent.

2. The load capacitance is defined as follows:

    (a) C = 100 ± 10 pF (including probe and parasitic capacitance) for sample dynamic tests.

    (b) C = 300 ± 30 pF (including probe and parasitic capacitance) for qualification dynamic tests.

    (c) The functional test shall be performed with only a 100 ± 10 pF load (including probe and parasitic capacitance) attached to each output.

    (d) C not required for any other tests.

FIGURE 9. Logic load circuit.

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NOTES: - Continued.

3. Select R2 for a current flow of Iol when point L is held at Vol. The values of Iol and Vol for each output are in the table below.

4. The value of R1 for each output load shall be as specified in the table below.

Pin number	R1 1/ (ohms)	Iol (mA)	Vol (volts)
4,6,13,14,22	6200	6.0	0.5
* 2,11,16,17,18,19, 23,24,27,28,29,32, 33,34,35,36,37,38, 39,41,42,43,44,45, 46,47,48,49,50,52, 53,54,55,56,57,58, 59,61,62,65,66,67, 68,69,70,71,72,73, 74,77,78,79,82,86, 87,88,91,92,93,94, 95,96,97,98,99,100	16000	1.6	0.5

1/ All resistors are metal film, ± 1 percent, 1/4 W.

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TABLE X. Output voltage, low level, test.

	Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
*	J1	EM-&-INT-PND	2	1.6	0.0	0.5	V	1-4
	J2	RESET-Y-FULL	4	6.0				
	J3	RST-MDRI-FUL	6	6.0				
	J4	SMI-D-IN-0	11	1.6				
	J5	WAIT-REQ	13	6.0				
	J6	READY	14	6.0				
	J7	PROM-47	16	1.6				
	J8	PROM-47	16					
	J9	PROM-46	17					
	J10	PROM-46	17					
*	J11	MCRO-P-10	18					
	J12	MCRO-P-11	19					
	J13	MCRO-P-11	19	1.6				
	J14	WAIT-ENABLE	22	6.0				
*	J15	SBEQADDGENSC	23	1.6				
*	J16	SB-EQ-ECI-SC	24					
	J17	MCROIR-45	27					
	J18	MCROIR-44	28					
	J19	MCROIR-43	29					
	J20	MCROIR-41	32					
	J21	MCROIR-40	33	1.6	0.0	0.5	V	1-4

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TABLE X. Output voltage, low level, test - Continued.

	Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
*	J22	XPANDED-ADRS	34	1.6	0.0	0.5	V	1-4
	J23	MCROIR-38	35					
	J24	MCROIR-37	36					
	J25	MCROIR-36	37					
	J26	MCROIR-35	38					
	J27	MCROIR-34	39					
	J28	MCROIR-33	41					
	J29	MCROIR-32	42					
	J30	MCROIR-31	43					
	J31	MCROIR-30	44					
	J32	MCROIR-29	45					
	J33	MCROIR-28	46					
	J34	MCROIR-27	47					
	J35	MCROIR-26	48					
	J36	MCROIR-25	49					
	J37	MCROIR-24	50					
*	J38	MICRO-P00	52					
*	J39	MICRO-P01	53					
*	J40	MICRO-P02	54					
*	J41	MICRO-P03	55					
*	J42	MICRO-P04	56	1.6	0.0	0.5	V	1-4

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TABLE X. Output voltage, low level, test - Continued.

	Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
*	J43	MICRO-P05	57	1.6	0.0	0.5	V	1-4
*	J44	MICRO-P06	58					
*	J45	MICRO-P07	59					
*	J46	MICRO-P08	61					
*	J47	MICRO-P09	62					
	J48	MCROIR-00	65					
	J49	MCROIR-01	66					
	J50	MCROIR-02	67					
	J51	MCROIR-03	68					
	J52	MCROIR-04	69					
	J53	MCROIR-05	70					
	J54	MCROIR-06	71					
	J55	MCROIR-07	72					
	J56	MCROIR-08	73					
	J57	MCROIR-09	74					
	J58	MCROIR-10	77					
	J59	MCROIR-11	78					
	J60	MCRO-CNT-FMT	79					
	J61	ECI-DESTNS	82					
*	J62	EMULATE-OE	86					
	J63	MCROIR-12	87	1.6	0.0	0.5	V	1-4

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TABLE X. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
J64	MCROIR-13	88	1.6	0.0	0.5	V	1-4
J65	MCROIR-14	91					
J66	MCROIR-15	92					
J67	MCROIR-16	93					
J68	MCROIR-17	94					
J69	MCROIR-18	95					
J70	MCROIR-19	96					
J71	MCROIR-20	97					
J72	MCROIR-21	98					
J73	MCROIR-22	99					
J74	MCROIR-23	100	1.6	0.0	0.5	V	1-4

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NOTES:

1. Output voltage, low level, test tolerance table.

	Condition	Value	Accuracy	Test condition
	Power supply, (Vcc)	4.5 V	± 45 mV	J1-J74
	Power supply current limit	3.82 A	NA	J1-J74
*	Forcing current, (Iol)	1.6 mA	± 160 µA	J1
*		6.0 mA	± 600 µA	J2,J3
		1.6 mA	± 160 µA	J4
		6.0 mA	± 600 µA	J5,J6
*		1.6 mA	± 160 µA	J7-J13
		6.0 mA	± 600 µA	J14
*		1.6 mA	± 160 µA	J15-J74
	Pattern voltage high, (VH)	3.0 V	± 100 mV	J1-J74
	low, (VL)	0.0 V	± 100 mV	J1-J74
	Output voltage measurement	NA	± 25 mV	J1-J74

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table X and the voltage then measured.

3. The load shall be as indicated on figure 9.

4. Abbreviations used are defined in 6.1.

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TABLE XI. Input clamp diode voltage test.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
L1	Y-FULL	3	-18.0	-1.2	0.0	V	1-3
L2	MDRI-FULL	5					
L3	PIPE-FULL	7					
L4	PAGING-BUSY	8					
L5	SET-BUS-REQ	9					
L6	SD-CLK	12					
L7	WAIT-ACK	15					
L8	CP-C4-CLK	20					
L9	CPSRC-A-B-EN	21		-1.2			
L10	SMI-DTA-OUT	26		-1.5			
L11	CPSCE-B-BS-E	30		-1.2			
L12	INTRPT-PNDNG	63					
L13	CP-C3-CLOCK	76					
L14	CP-MACLR	80					
L15	MP-SYS-CLK	84					
L16	SMI-LD-CTRL	85					
L17	CP-SCE-B-CLK	89	-18.0	-1.2	0.0	V	1-3

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NOTES:

1. Input clamp diode voltage test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	OFF	NA	L1-L17
Power supply current limit	NA	NA	L1-L17
Forcing current, (Iin)	-18 mA	± 1.8 mA	L1-L17
Pattern voltage high, (VH) low, (VL)	NA NA	NA NA	L1-L17 L1-L17
Input voltage measurement	NA | NA	± 60 mV ± 80 mV ± 60 mV	L1-L9 L10 L11-L17

2. The following procedure is required to perform this test:

(a) No power shall be applied to the module.

(b) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(c) The forcing current shall be applied to the pin specified in the appropriate test conditions in table XI and the voltage then measured.

3. Abbreviations used are defined in 6.1.

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TABLE XII. Power supply filter capacitance test.

Test cond	Parameter	Pin numbers	25 ± 5°C Initial limits Min	Max	Units	Notes
M1	PSFC	1	4.9	9.1	μF	1-4
M2	PSFC	51	5.6	10.4	μF	1-4

NOTES:

1. Power supply filter capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance bridge frequency	1 KHz	± 100 Hz	M1,M2
Capacitance bridge voltage	0.3 Vp	± 30 mVp	M1,M2
Capacitance measurement	NA NA	± 210 nF ± 240 nF	M1 M2

2. The capacitance bridge used in this test shall be item C01. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XII.

3. The test circuit shall be as indicated on figure 10.

4. Abbreviations used are defined in 6.1.

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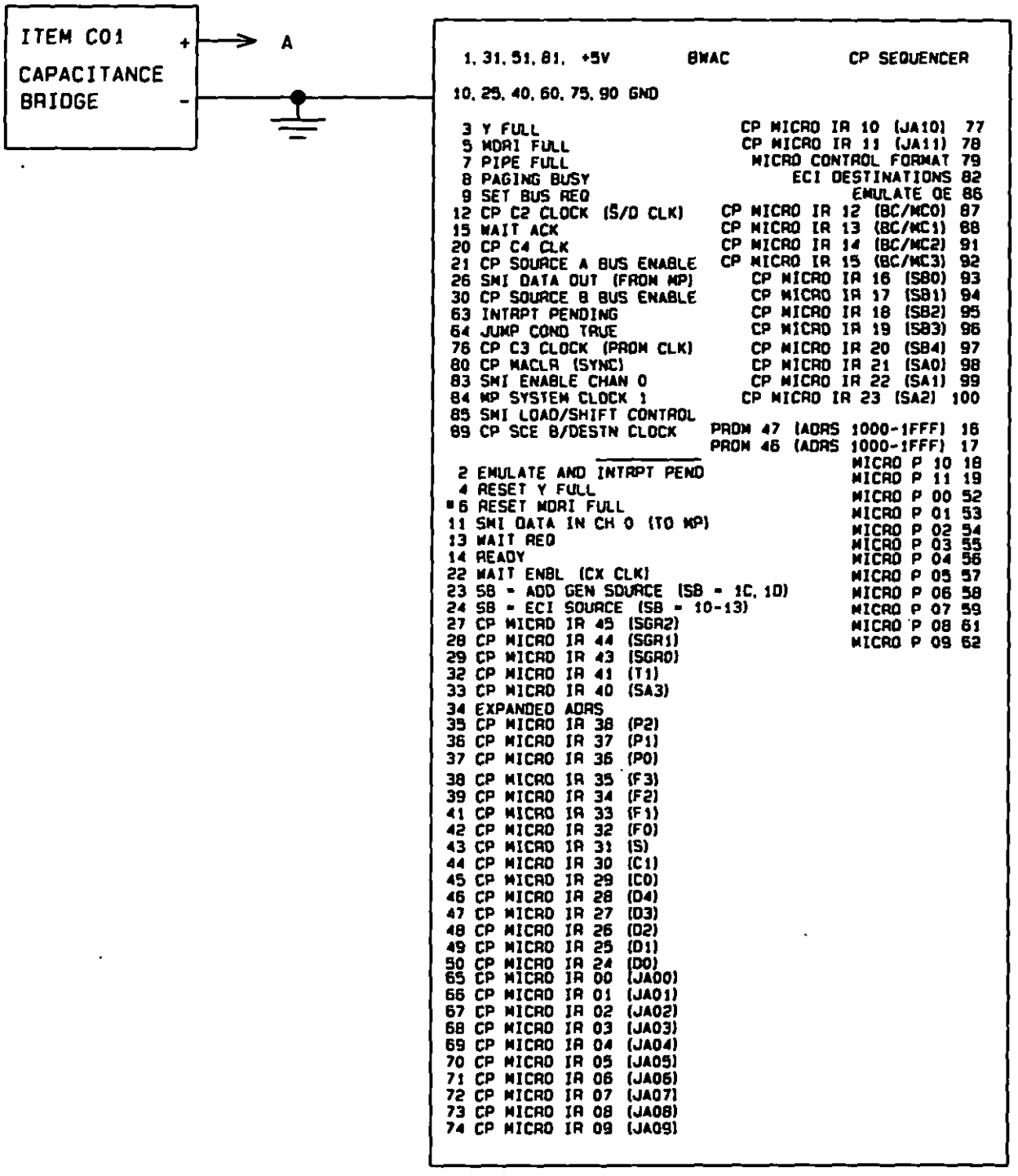


FIGURE 10. Power supply filter capacitance test circuit for table XII.

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TABLE XIII. Power supply current test.

	Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
*	N	Icc	1,31, 51,81	630	2544	mA	1-3

NOTES:

1. Power supply current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	N
Power supply current limit	3.82 A	NA	N
Pattern voltage high, (VH)	3.0 V	± 100 mV	N
low, (VL)	0.0 V	± 100 mV	N
* Power supply current measurement	NA	± 95.7 mA	N

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test and take a measurement at the pattern line with the maximum power supply current as indicated in table XIII.

3. Abbreviations used are defined in 6.1.

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TABLE XIV. Functional test.

Test cond	Test description	Notes
P1	Functional test minimum power supply voltage maximum pattern rate (225 nS)	1-3
P2	Functional test maximum power supply voltage maximum pattern rate (225 nS)	1-3

NOTES:

1. Functional test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	P1 P2
Power supply current limit	3.82 A	NA	P1,P2
Pattern voltage high, (VH) low, (VL)	 2.4 V 0.4 V	 ± 100 mV ± 100 mV	 P1,P2 P1,P2
Output voltage verification high, (Voh) high, (Voh) low, (Vol)	 NA NA	 ± 105 mV ± 155 mV ± 25 mV	 P1 P2 P1,P2

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Use the appropriate pattern to verify the functional performance of the module as indicated in table XIV using only a capacitive load of 100 pF (see figure 9). The output compare levels shall be as follows:

High level: $2.4\ V \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq 0.5\ V$

(c) The module inputs shall not violate the minimum input timing parameters referenced on figure 11.

3. Abbreviations used are defined in 6.1.

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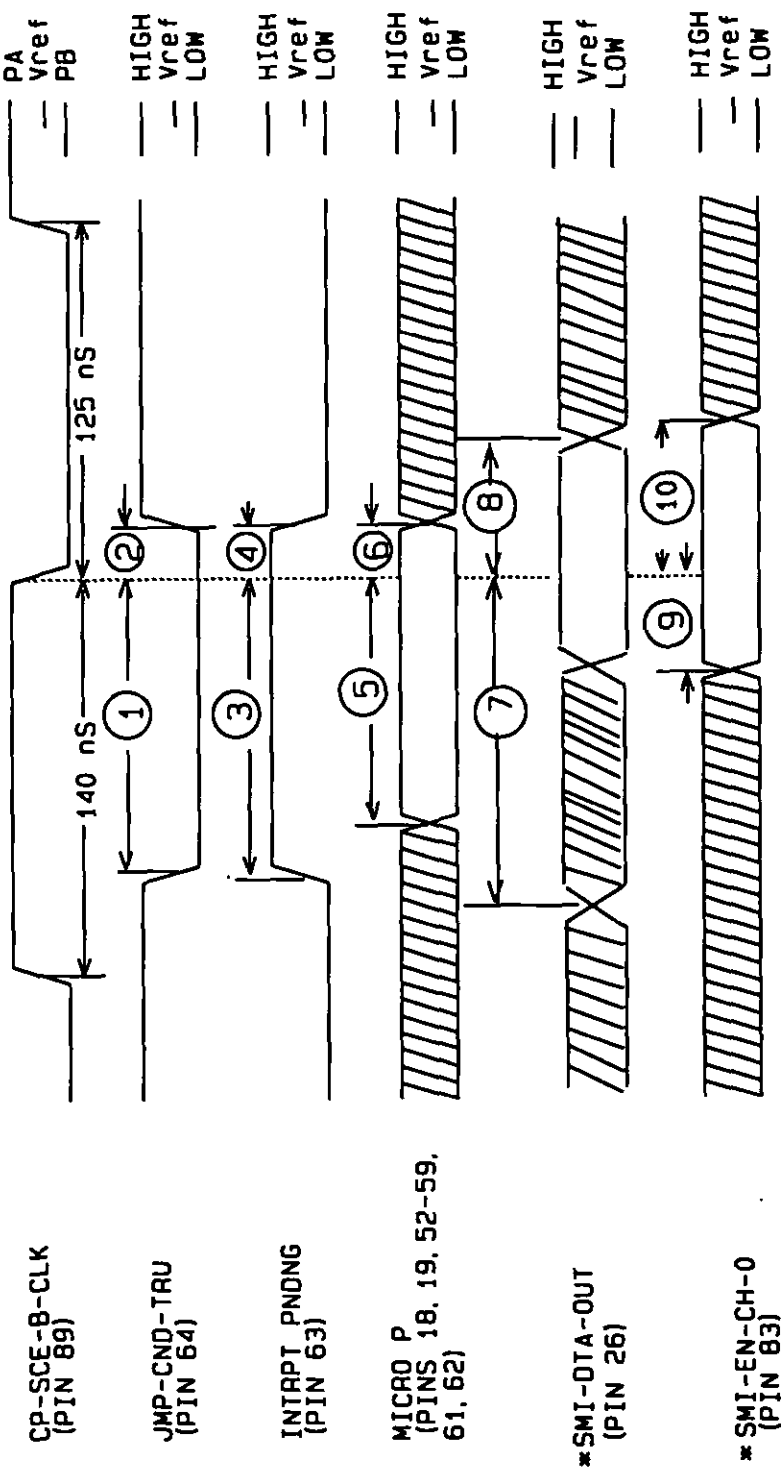


FIGURE 11. Input timing parameters for table XIV.

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NOTES:

1. (1) JMP-CND-TRU to CP-SCE-B-CLK. Setup time: minimum 125 nS.

2. (2) CP-SCE-B-CLK to JMP-CND-TRU. Hold time: minimum 10 nS.

3. (3) INTRPT-PNDNG to CP-SCE-B-CLK. Setup time: minimum 125 nS.

4. (4) CP-SCE-B-CLK to INTRPT-PNDNG. Hold time: minimum 10 nS.

5. (5) MICRO P to CP-SCE-B-CLK. Setup time: minimum 85 nS.

6. (6) CP-SCE-B-CLK to MICRO P. Hold time: minimum 10 nS.

7. (7) SMI-DTA-OUT to CP-SCE-B-CLK. Setup time: minimum 25 nS.

*8. (8) CP-SCE-B-CLK to SMI-DTA-OUT. Hold time: minimum 30 nS.

*9. (9) SMI-EN-CH-0 to CP-SCE-B-CLK. Setup time: minimum 153 nS.

*10. (10) CP-SCE-B-CLK to SMI-EN-CH-0. Hold time: minimum 30 nS.

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TABLE XV. Dynamic test.

Test cond	Test description	Notes
Q1	Dynamic test, sample maximum pattern rate (225 nS) minimum power supply	1-6
Q2	Dynamic test, sample maximum pattern rate (225 nS) maximum power supply	1-6
Q3	Dynamic test, qualification maximum pattern rate (225 nS) minimum power supply	1-6
Q4	Dynamic test, qualification maximum pattern rate (225 nS) maximum power supply	1-6

NOTES:

1. Dynamic test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	Q1,Q3 Q2,Q4
Power supply current limit	3.82	NA	Q1-Q4
Input voltage high, (VH) low, (VL)	 2.4 V 0.4 V	 ± 100 mV ± 100 mV	 Q1-Q4 Q1-Q4

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Output voltage verification			
high, (Voh)	NA	± 100 mV	Q1,Q3
high, (Voh)	|	|	Q2,Q4
low, (Vol)	NA	± 100 mV	Q1-Q4

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

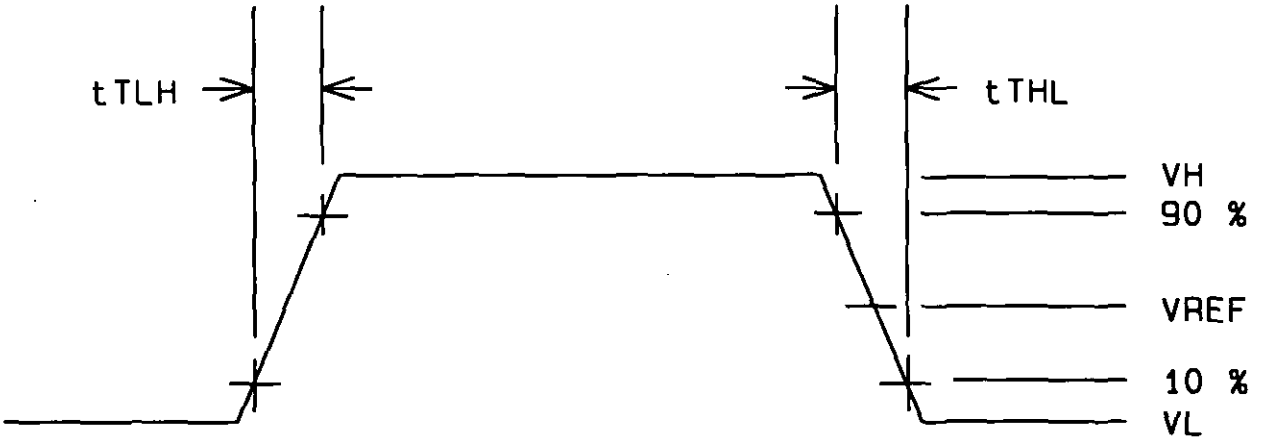
(b) The output compare levels shall be as follows:

High level: $Vref \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq Vref$

(c) The dynamic parameters are found in table XVI.

3. The load circuit for this test is specified on figure 9.

4. Input/Output (I/O) waveforms are as referenced on figure 12. All of the input waveforms shall have the following characteristics:



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NOTES: - Continued.

Where:

(a) VH and VL are as specified in the test tolerance table.

(b) tTLH = tTHL $\leq$ 10 nS.

(c) Vref = 1.6 V.

5. This test shall be performed as indicated in table XV.

6. Abbreviations used are defined in 6.1.

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TABLE XVI. Dynamic parameters.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1	tpd	CP-SCE-B-CLK to MCROIR-45	-	-	-	50	50	50	nS	12
		CP-SCE-B-CLK to MCROIR-44	-	-	-					
		CP-SCE-B-CLK to MCROIR-43	-	-	-					
		CP-SCE-B-CLK to MCROIR-41	-	-	-					
		CP-SCE-B-CLK to MCROIR-40	-	-	-					
		CP-SCE-B-CLK to MCROIR-38	-	-	-					
		CP-SCE-B-CLK to MCROIR-37	-	-	-					
		CP-SCE-B-CLK to MCROIR-36	-	-	-					
		CP-SCE-B-CLK to MCROIR-35	-	-	-					
		CP-SCE-B-CLK to MCROIR-34	-	-	-					
		CP-SCE-B-CLK to MCROIR-33	-	-	-					
		CP-SCE-B-CLK to MCROIR-32	-	-	-					
		CP-SCE-B-CLK to MCROIR-31	-	-	-					
1	tpd	CP-SCE-B-CLK to MCROIR-30	-	-	-	50	50	50	nS	12

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TABLE XVI. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1	tpd	CP-SCE-B-CLK to MCROIR-29	-	-	-	50	50	50	nS	12
		CP-SCE-B-CLK to MCROIR-28	-	-	-					
		CP-SCE-B-CLK to MCROIR-27	-	-	-					
		CP-SCE-B-CLK to MCROIR-26	-	-	-					
		CP-SCE-B-CLK to MCROIR-25	-	-	-					
		CP-SCE-B-CLK to MCROIR-24	-	-	-					
		CP-SCE-B-CLK to MCROIR-00	-	-	-					
		CP-SCE-B-CLK to MCROIR-01	-	-	-					
		CP-SCE-B-CLK to MCROIR-02	-	-	-					
		CP-SCE-B-CLK to MCROIR-03	-	-	-					
		CP-SCE-B-CLK to MCROIR-04	-	-	-					
		CP-SCE-B-CLK to MCROIR-05	-	-	-					
		CP-SCE-B-CLK to MCROIR-06	-	-	-					
1	tpd	CP-SCE-B-CLK to MCROIR-07	-	-	-	50	50	50	nS	12

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TABLE XVI. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1	tpd	CP-SCE-B-CLK to MCROIR-08	-	-	-	50	50	50	nS	12
		CP-SCE-B-CLK to MCROIR-09	-	-	-					
		CP-SCE-B-CLK to MCROIR-10	-	-	-					
		CP-SCE-B-CLK to MCROIR-11	-	-	-					
		CP-SCE-B-CLK to MCROIR-12	-	-	-					
		CP-SCE-B-CLK to MCROIR-13	-	-	-					
		CP-SCE-B-CLK to MCROIR-14	-	-	-					
		CP-SCE-B-CLK to MCROIR-15	-	-	-					
		CP-SCE-B-CLK to MCROIR-16	-	-	-					
		CP-SCE-B-CLK to MCROIR-17	-	-	-					
		CP-SCE-B-CLK to MCROIR-18	-	-	-					
		CP-SCE-B-CLK to MCROIR-19	-	-	-					
		CP-SCE-B-CLK to MCROIR-20	-	-	-					
1	tpd	CP-SCE-B-CLK to MCROIR-21	-	-	-	50	50	50	nS	12

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TABLE XVI. Dynamic parameters - Continued.

	Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
				-55°C	25°C	85°C	-55°C	25°C	85°C		
	1	tpd	CP-SCE-B-CLK to MCROIR-22	-	-	-	50	50	50	nS	12
	1		CP-SCE-B-CLK to MCROIR-23	-	-	-	50	50	50		
	2		CP-SCE-B-CLK to RST-MDRI-FULL	-	-	-	35	35	35		
	2		CP-SCE-B-CLK to RESET-Y-FULL	-	-	-	35	35	35		
*	3		CP-SCE-B-CLK to MCRO-CNT-FMT	-	-	-	50	50	50		
	4		CP-SCE-B-CLK to SBEQADDGENSC	-	-	-	65	65	65		
	4		CP-SCE-B-CLK to SB-EQ-ECI-SC	-	-	-	65	65	65		
	5		CP-SCE-B-CLK to EMULATE-OE	-	-	-	85	85	85		
			CP-SCE-B-CLK to WAIT-REQ	-	-	-					
	5		CP-SCE-B-CLK to READY	-	-	-	85	85	85		
	6		CP-SCE-B-CLK to EM-&-INT-PND	-	-	-	145	145	145		
	6	tpd	CP-SCE-B-CLK to MCRO-P-10	-	-	-	145	145	145	nS	12

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TABLE XVI. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
6	tpd	CP-SCE-B-CLK to MCRO-P-11	-	-	-	145	145	145	nS	12
		CP-SCE-B-CLK to MCRO-P00	-	-	-					
		CP-SCE-B-CLK to MCRO-P01	-	-	-					
		CP-SCE-B-CLK to MCRO-P02	-	-	-					
		CP-SCE-B-CLK to MCRO-P03	-	-	-					
		CP-SCE-B-CLK to MCRO-P04	-	-	-					
		CP-SCE-B-CLK to MCRO-P05	-	-	-					
		CP-SCE-B-CLK to MCRO-P06	-	-	-					
		CP-SCE-B-CLK to MCRO-P07	-	-	-					
		CP-SCE-B-CLK to MCRO-P08	-	-	-					
6		CP-SCE-B-CLK to MCRO-P09	-	-	-	145	145	145		
7		CP-SCE-B-CLK to PROM-47	-	-	-	225	225	225		
7		CP-SCE-B-CLK to PROM-46	-	-	-	225	225	225		
8	tpd	Y-FULL to WAIT-REQ	-	-	-	25	25	25	nS	12

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TABLE XVI. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
8	tpd	MDRI-FULL to WAIT-REQ	-	-	-	25	25	25	nS	12
		PIPE-FULL to WAIT-REQ	-	-	-					
		PAGING BUSY to WAIT-REQ	-	-	-					
		SET-BUS-REQ to WAIT-REQ	-	-	-					
8		SD-CLK to WAIT-REQ	-	-	-					
9		WAIT-ACK to READY	-	-	-					
10		SD-CLK to WAIT-ENABLE	-	-	-					
11		CP-C4-CLK to WAIT-ENABLE	-	-	-	25	25	25		
12	tpd	MP-SYS-CLK to SMI-D-IN-0	-	-	-	60	60	60		
13	tsu	JMP-CND-TRU to CP-SCE-B-CLK	125	125	125	-	-	-		
14	th	CP-SCE-B-CLK to JMP-CND-TRU	10	10	10	-	-	-		
15	tsu	INTRPT-PNDNG to CP-SCE-B-CLK	125	125	125	-	-	-	nS	12

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TABLE XVI. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
16	th	CP-SCE-B-CLK to INTRPT-PNDNG	10	10	10	-	-	-	nS	12
17	tsu	MCRO-P-10 to CP-SCE-B-CLK	85	85	85	-	-	-		
		MCRO-P-11 to CP-SCE-B-CLK				-	-	-		
		MICRO-P00 to CP-SCE-B CLK				-	-	-		
		MICRO-P01 to CP-SCE-B-CLK				-	-	-		
		MICRO-P02 to CP-SCE-B-CLK				-	-	-		
		MICRO-P03 to CP-SCE-B-CLK				-	-	-		
		MICRO-P04 to CP-SCE-B-CLK				-	-	-		
		MICRO-P05 to CP-SCE-B-CLK				-	-	-		
17	tsu	MICRO-P06 to CP-SCE-B-CLK	85	85	85	-	-	-	nS	12

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TABLE XVI. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
17	tsu	MICRO-P07 to CP-SCE-B-CLK	85	85	85	-	-	-	nS	12
		MICRO-P08 to CP-SCE-B-CLK				-	-	-		
17	tsu	MICRO-P09 to CP-SCE-B-CLK	85	85	85	-	-	-		
18	th	CP-SCE-B-CLK to MCRO-P-10	10	10	10	-	-	-		
		CP-SCE-B-CLK to MCRO-P-11				-	-	-		
		CP-SCE-B-CLK to MICRO-P00				-	-	-		
		CP-SCE-B-CLK to MICRO-P01				-	-	-		
		CP-SCE-B-CLK to MICRO-P02				-	-	-		
		CP-SCE-B-CLK to MICRO-P03				-	-	-		
		CP-SCE-B-CLK to MICRO-P04				-	-	-		
		CP-SCE-B-CLK to MICRO-P05				-	-	-		
		CP-SCE-B-CLK to MICRO-P06				-	-	-		
18	th	CP-SCE-B-CLK to MICRO-P07	10	10	10	-	-	-	nS	12

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TABLE XVI. Dynamic parameters - Continued.

	Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
				-55°C	25°C	85°C	-55°C	25°C	85°C		
	18	th	CP-SCE-B-CLK to MICRO-P08	10	10	10	-	-	-	nS	12
	18	th	CP-SCE-B-CLK to MICRO-P09	10	10	10	-	-	-		
	19	tsu	SMI-DTA-OUT to CP-SCE-B-CLK	25	25	25	-	-	-		
	20	th	CP-SCE-B-CLK to SMI-DTA-OUT	30	30	30	-	-	-		
*	21	tsu	SMI-EN-CH-0 to CP-SCE-B-CLK	153	153	153	-	-	-		
*	22	th	CP-SCE-B-CLK to SMI-EN-CH-0	30	30	30	-	-	-	nS	12

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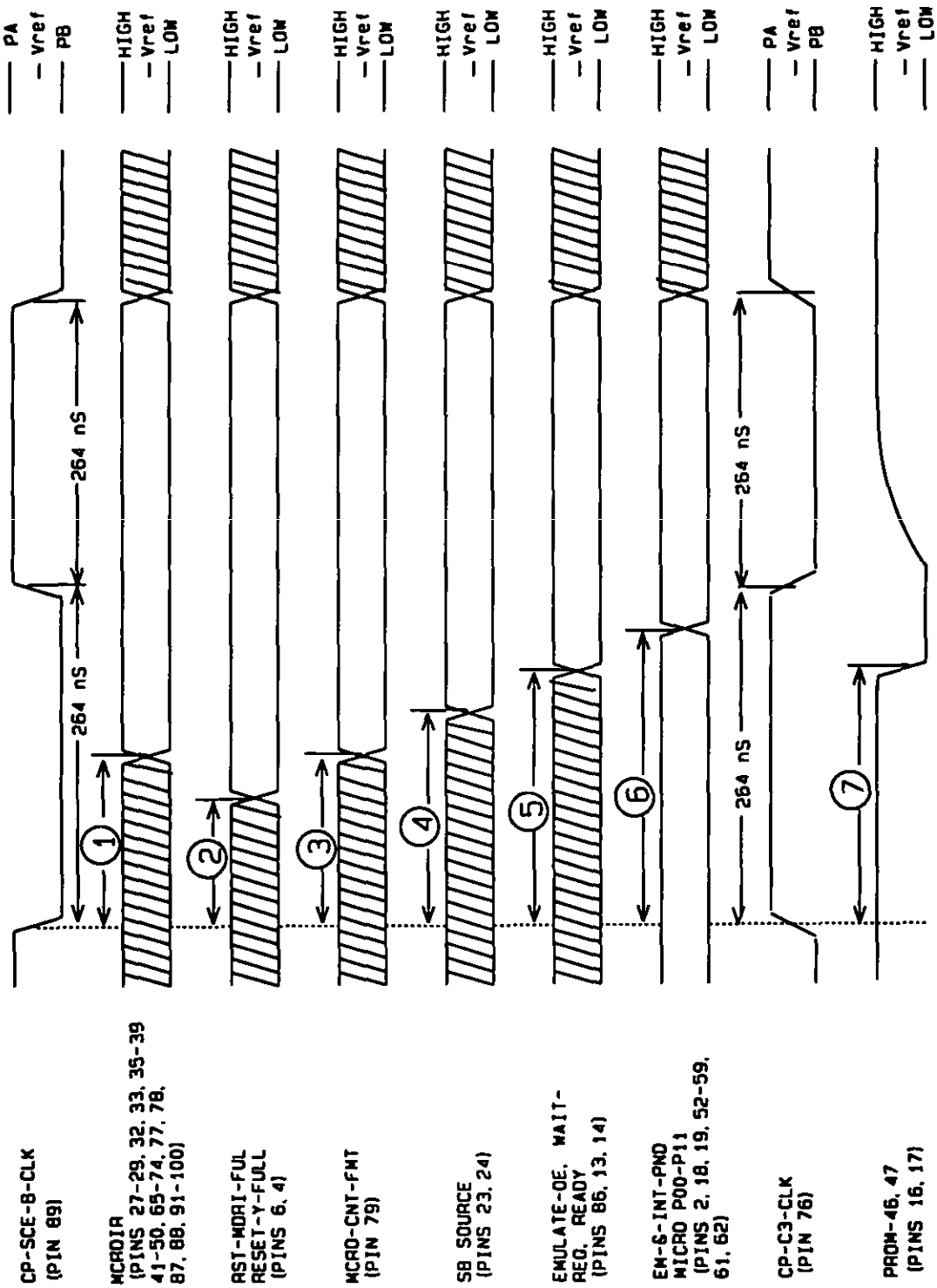


FIGURE 12a. Dynamic waveforms for table XVI.

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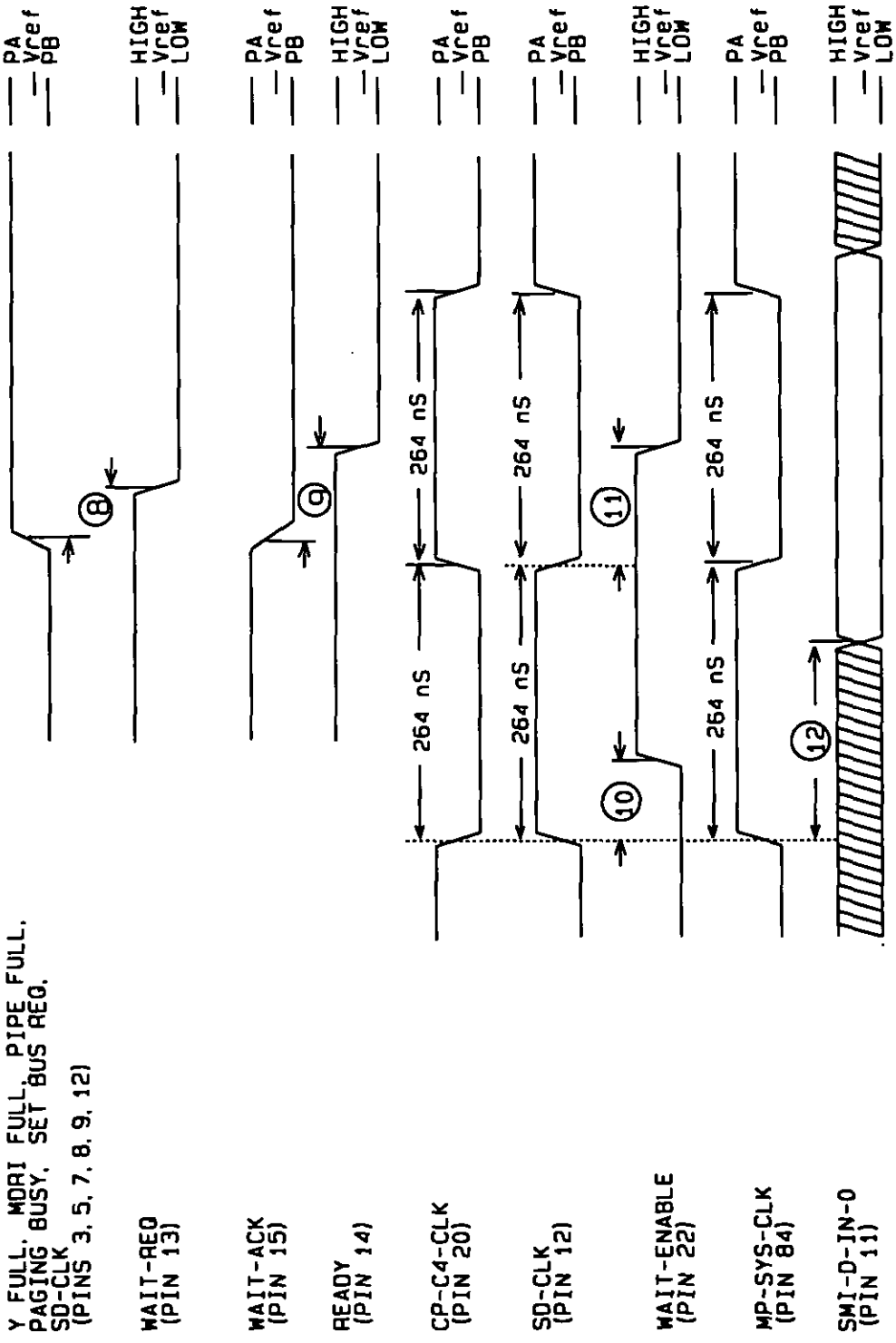


FIGURE 12b. Dynamic waveforms for table XVI.

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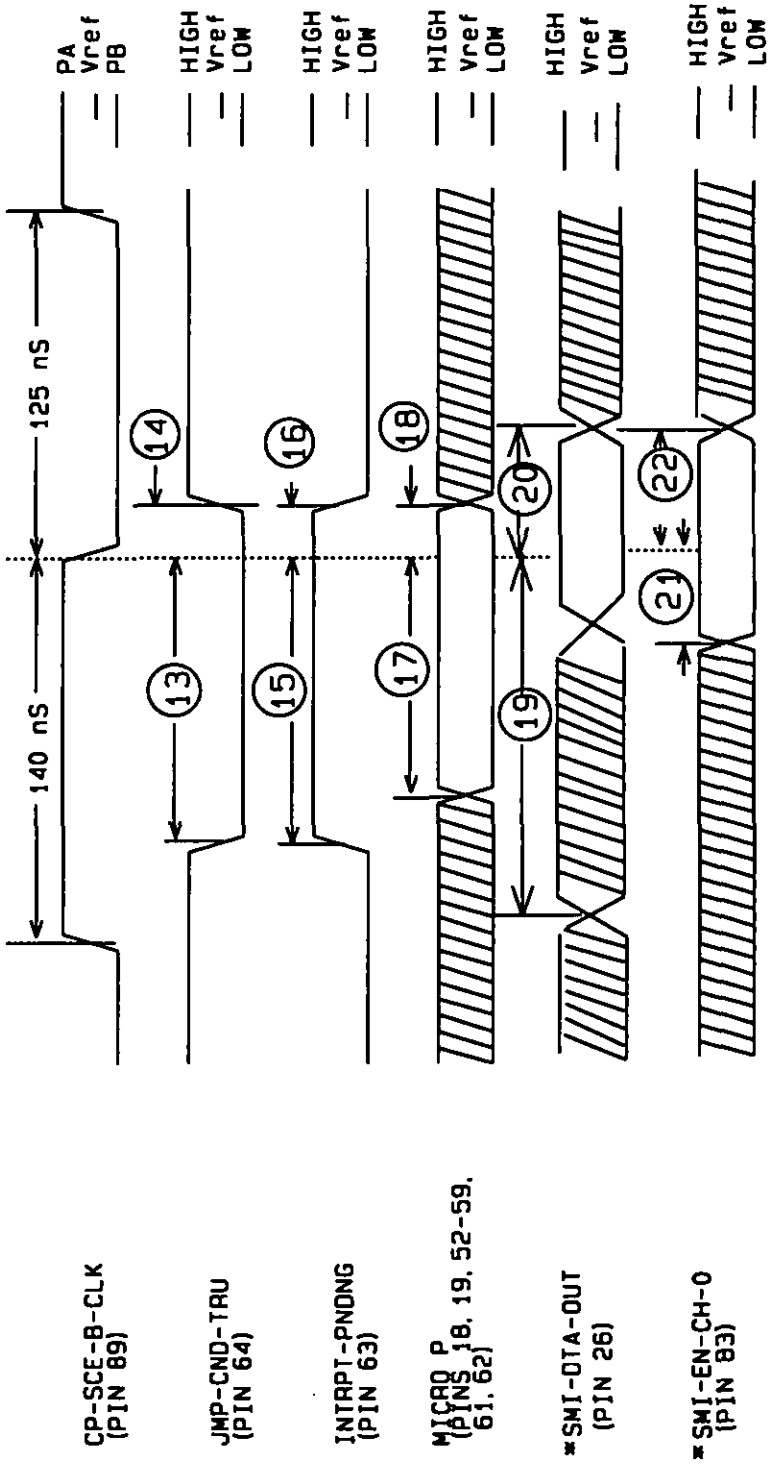


FIGURE 12c. Dynamic waveforms for table XVI.

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TABLE XVII. Input and output capacitance test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
Inputs						
R1	Y-FULL	3	2	50	pF	1-4
R2	MDRI-FULL	5				
R3	PIPE-FULL	7				
R4	PAGING-BUSY	8				
R5	SET-BUS-REQ	9	2	50		
R6	SD-CLK	12	10	100		
R7	WAIT-ACK	15	2	50		
R8	CP-C4-CLK	20				
R9	CPSRC-A-B-EN	21				
R10	SMI-DTA-OUT	26				
R11	CPSCE-B-BS-E	30	2	50		
R12	INTRPT-PNDNG	63	4	75		
R13	JMP-CND-TRU	64	8	175		
R14	CP-C3-CLK	76	4	100		
R15	CP-MACLR	80	2	50		
R16	SMI-EN-CH-0	83	8	125		
R17	MP-SYS-CLK	84	4	75		
R18	SMI-LD-CTRL	85	2	75		
R19	CP-SCE-B-CLK	89	6	100	pF	1-4

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TABLE XVII. Input and output capacitance test - Continued.

	Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
	Bidirectionals						
	R20	MCRO-P-10	18	16	321	pF	1-4
	R21	MCRO-P-11	19	16	312		
	R22	MICRO-P00	52	14	364		
*	R23	MICRO-P01	53		200		
	R24	MICRO-P02	54		273		
	R25	MICRO-P03	55		263		
	R26	MICRO-P04	56		262		
	R27	MICRO-P05	57		254		
	R28	MICRO-P06	58		261		
	R29	MICRO-P07	59		283		
	R30	MICRO-P08	61		279		
	R31	MICRO-P09	62	14	328		
	Outputs						
	R32	MCROIR-45	27	2	50		
	R33	MCROIR-44	28	2	50		
	R34	MCROIR-43	29	4	75		
	R35	MCROIR-41	32		75		
	R36	MCROIR-40	33	4	100		
	R37	XPANDED-ADRS	34	12	204		
	R38	MCROIR-38	35	10	150	pF	1-4

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TABLE XVII. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
R39	MCROIR-37	36	8	175	pF	1-4
R40	MCROIR-36	37	8	200		
R41	MCROIR-35	38	2	50		
R42	MCROIR-34	39				
R43	MCROIR-33	41				
R44	MCROIR-32	42				
R45	MCROIR-31	43				
R46	MCROIR-30	44				
R47	MCROIR-29	45	2	50		
R48	MCROIR-28	46	4	75		
R49	MCROIR-27	47				
R50	MCROIR-26	48				
R51	MCROIR-25	49				
R52	MCROIR-24	50				
R53	MCROIR-00	65				
R54	MCROIR-01	66				
R55	MCROIR-02	67				
R56	MCROIR-03	68				
R57	MCROIR-04	69				
R58	MCROIR-05	70				
R59	MCROIR-06	71	4	75	pF	1-4

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TABLE XVII. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
R60	MCROIR-07	72	4	75	pF	1-4
R61	MCROIR-08	73				
R62	MCROIR-09	74				
R63	MCROIR-10	77				
R64	MCROIR-11	78	4			
R65	MCROIR-12	87	2	75		
R66	MCROIR-13	88		50		
R67	MCROIR-14	91	2	75		
R68	MCROIR-16	93	4	150		
R69	MCROIR-17	94	6	200		
R70	MCROIR-18	95	4	75		
R71	MCROIR-19	96	2			
R72	MCROIR-20	97	4			
R73	MCROIR-21	98				
R74	MCROIR-22	99				
R75	MCROIR-23	100	4			
R76	MCROIR-15	92	2	75	pF	1-4

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NOTES:

1. Input and output capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance meter frequency	1 MHz	± 100 KHz	R1-R76
Capacitance meter voltage	15 mV	± 1.5 mV	R1-R76
Capacitance	NA	± 2.4 pF	R1-R5
		± 4.5 pF	R6
		± 2.4 pF	R7-R11
		± 3.55 pF	R12
		± 8.35 pF	R13
		± 4.8 pF	R14
		± 2.4 pF	R15
		± 5.85 pF	R16
		± 3.55 pF	R17
		± 3.65 pF	R18
		± 4.7 pF	R19
		± 15.3 pF	R20
		± 14.8 pF	R21
		± 17.5 pF	R22
		± 9.3 pF	R23
		± 13.0 pF	R24
		± 12.5 pF	R25
		± 12.4 pF	R26
		± 12.0 pF	R27
		± 12.4 pF	R28
		± 13.5 pF	R29
		± 13.3 pF	R30
	NA	± 15.7 pF	R31

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Capacitance	NA	± 2.4 pF	R32,R33
		± 3.55 pF	R34,R35
		± 4.8 pF	R36
		± 9.6 pF	R37
		± 7.0 pF	R38
		± 8.35 pF	R39
		± 9.6 pF	R40
		± 2.4 pF	R41-R47
		± 3.55 pF	R48-R64
		± 3.65 pF	R65
		± 2.4 pF	R66
		± 3.65 pF	R67
		± 7.3 pF	R68
		± 9.7 pF	R69
		± 3.55 pF	R70
		± 3.65 pF	R71
		± 3.55 pF	R72-R75
	NA	± 3.65 pF	R76

2. The capacitance meter used in this test shall be item C05. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XVII.

3. The test circuit shall be as indicated on figure 13.

4. Abbreviations used are defined in 6.1.

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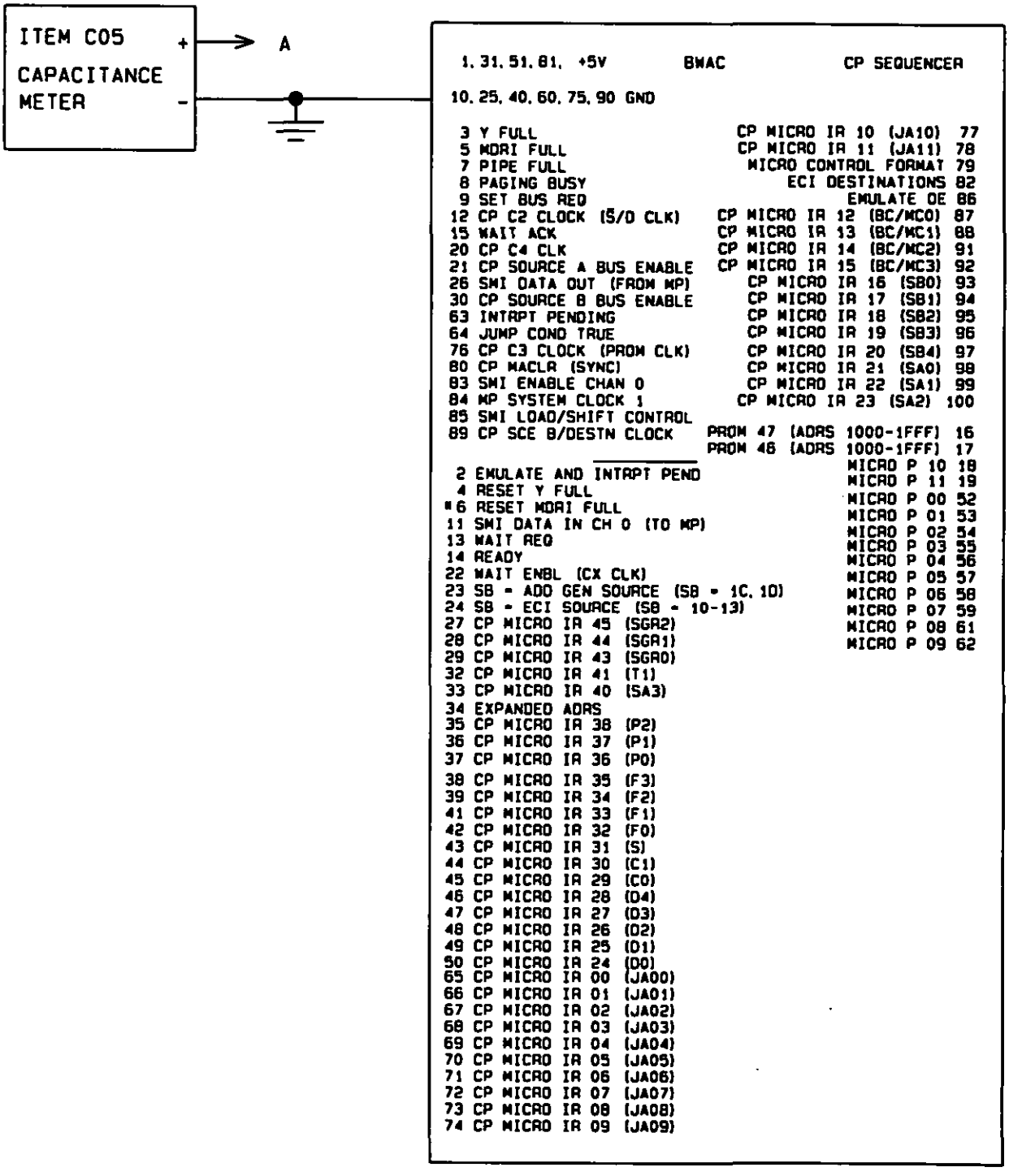


FIGURE 13. Input and output capacitance test circuit for table XVII.

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TABLE XVIII. Power supply transient amplitude test.

Test cond	Parameter	Pin number	25 ± 5°C Initial limits Min	Max	Units	Notes
T1	PSTA	1	0	400	mV	1-4
T2	PSTA	51	0	400	mV	1-4

NOTES:

1. Power supply transient amplitude test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 250 mV	T1,T2
Power supply current limit	3.82 A	NA	T1,T2
Pattern voltage high, (VH)	3.0 V	± 100 mV	T1,T2
low, (VL)	0.0 V	± 100 mV	T1,T2
Power supply transient amplitude measurement	NA	± 20 mV	T1,T2

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table to an external source. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test and make the indicated measurement for the test condition shown in table XVIII.

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NOTES: - Continued.

(c) Vcc pins 1 and 31 shall be shorted together, pins 51 and 81 shall be shorted together and the scope point, probe 'A', shall be connected to the pin indicated in table XVIII.

(d) Using the oscilloscope, D06, inspect for the most negative spike as shown on figure 14 at the pattern line with the maximum power supply current.

(e) The maximum probe capacitance is 15 pF. There shall be no other capacitance present on Vcc to ground.

3. The test circuit shall be as indicated on figure 15.

4. Abbreviations used are defined in 6.1.

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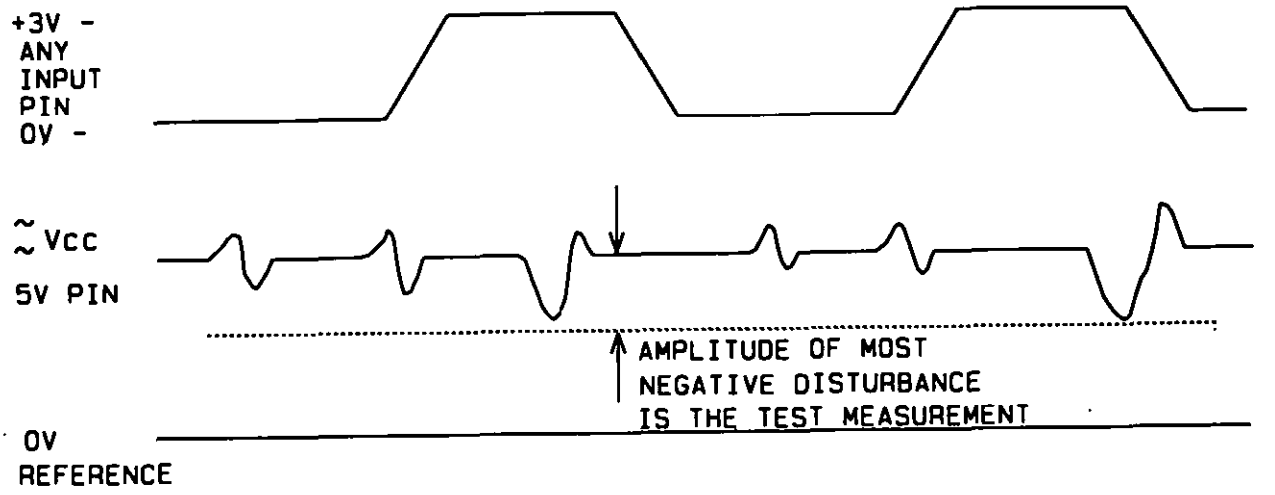
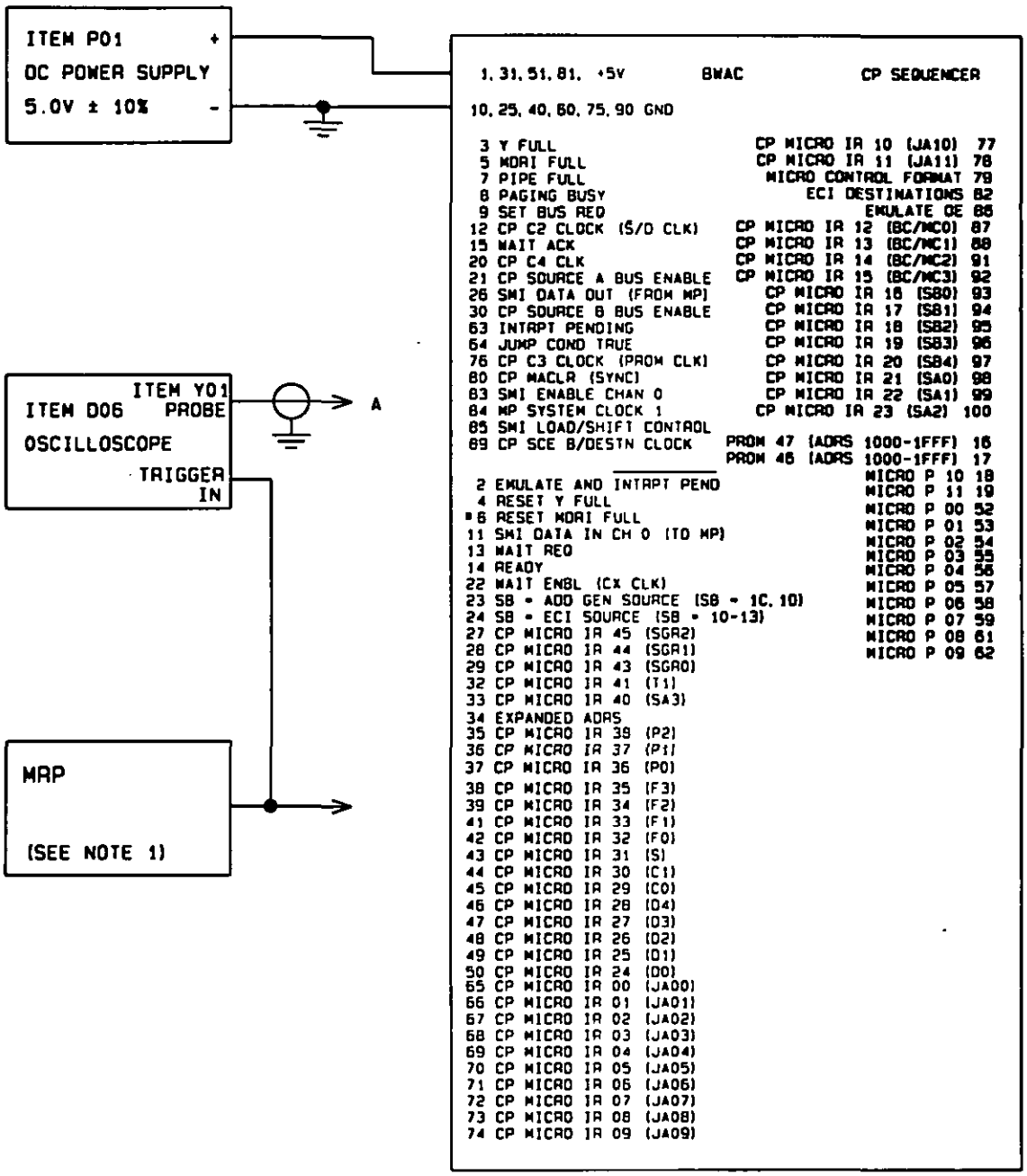


FIGURE 14. Power supply transient amplitude waveform for table XVIII.

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NOTES:

1. MRP or emulator capable of reproducing system backpanel characteristics.

FIGURE 15. Power supply transient amplitude test circuit for table XVIII.

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	MIL-M-28787/508A	1994 March 23

3. DOCUMENT TITLE
Modules, Standard Electronic, Central Processor Sequencer Module, Key Code BWAC

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

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a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
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a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
Technical Point of Contact (TPOC): Mr. Joseph P. Toy (Code 603A) PLEASE ADDRESS ALL CORRESPONDENCE AS FOLLOWS:	TPOC: 812-854-5993	88-482-5993
c. ADDRESS (Include Zip Code) Commander ATTN: Joseph P. Toy, Code 603A, Bldg. 2917 NAVSURFWARCENDIV, 300 Highway 361 Crane, IN 47522-5001	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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